Chapter 10:Gibbs Free Energy Composition and Phase Diagrams of Binary Systems

Part III

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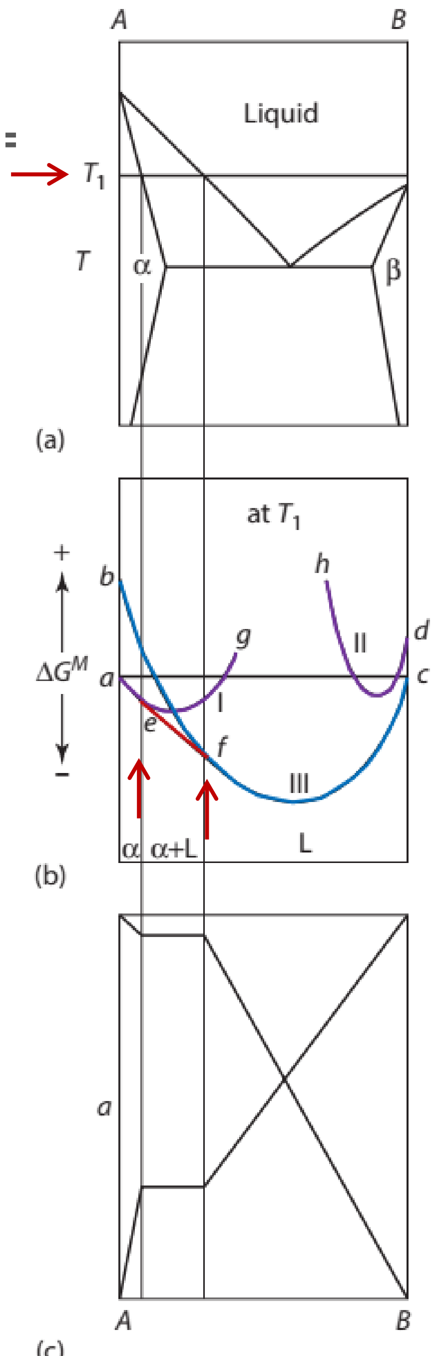
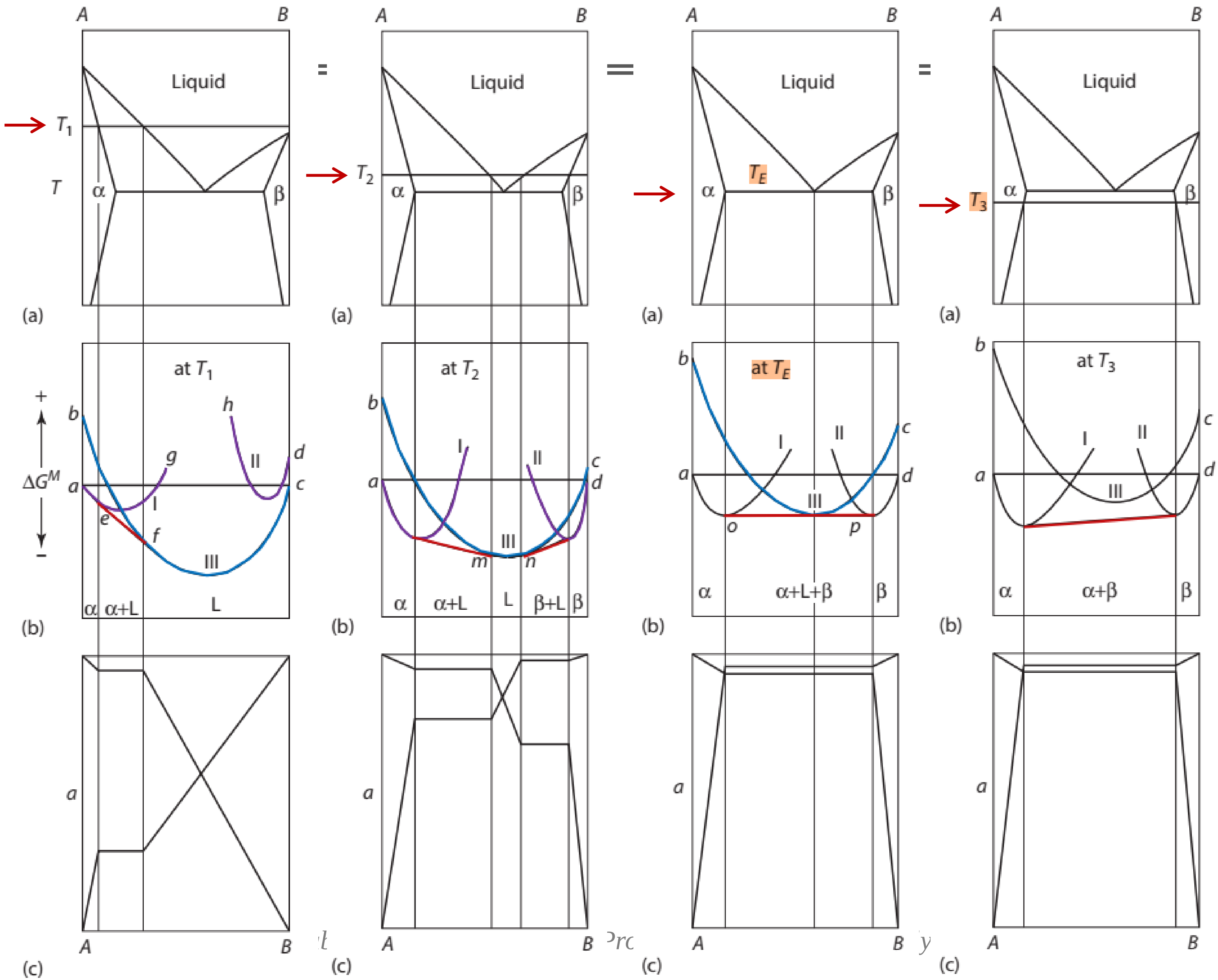


10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity

- ✓ Complete mutual solid solubility of the components *A* and *B* requires that *A* and *B* have the same crystal structures, that they be of comparable atomic size, and that they have similar electronegativities and valencies.

- ✓ If any one of these conditions is not met, then one or more two-phase regions will **occur** in the solid state.

Figure 10.15 the effect of temperature on the molar Gibbs free energies of mixing and the activities of the components of the system *A*–*B*.



$T\downarrow$

T_1

↓

T_2

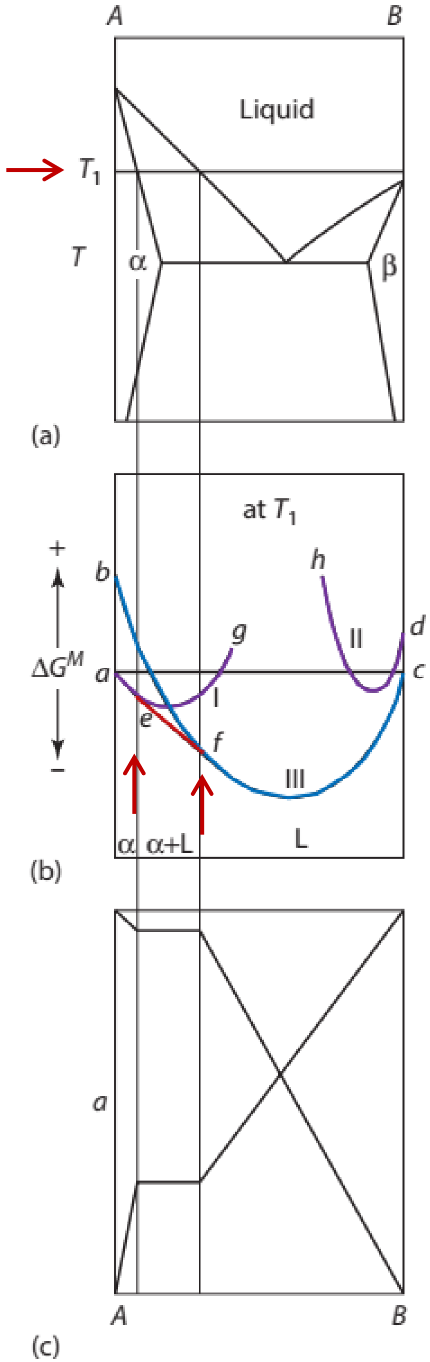
↓

T_E

↓

T_3

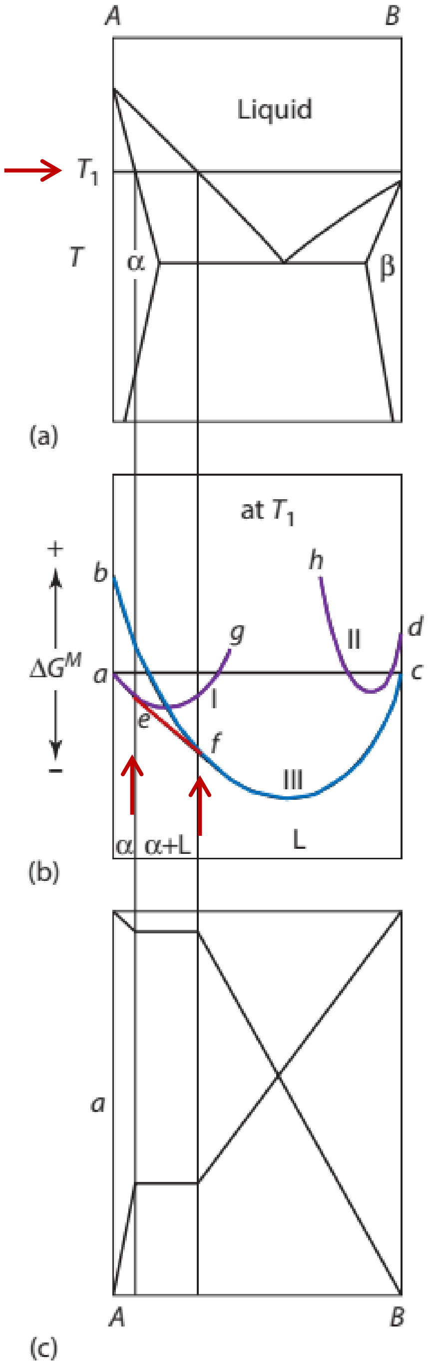
10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity



- ✓ Consider the **eutectic *A–B*** binary system, the phase diagram of which is shown in Figure 10.15a, in which *A* and *B* have different crystal structures. Two terminal solid solutions, ***α*** and ***β***, occur.
- ✓ Figure 10.15b: the molar ΔG^M curves at T_1
- ✓ Figure 10.15c shows the a–X_B relationships of the components at T_1, drawn with respect to Solid as the standard state for *A* (point ***a***) and Liquid as the standard state for *B* (point ***c***).

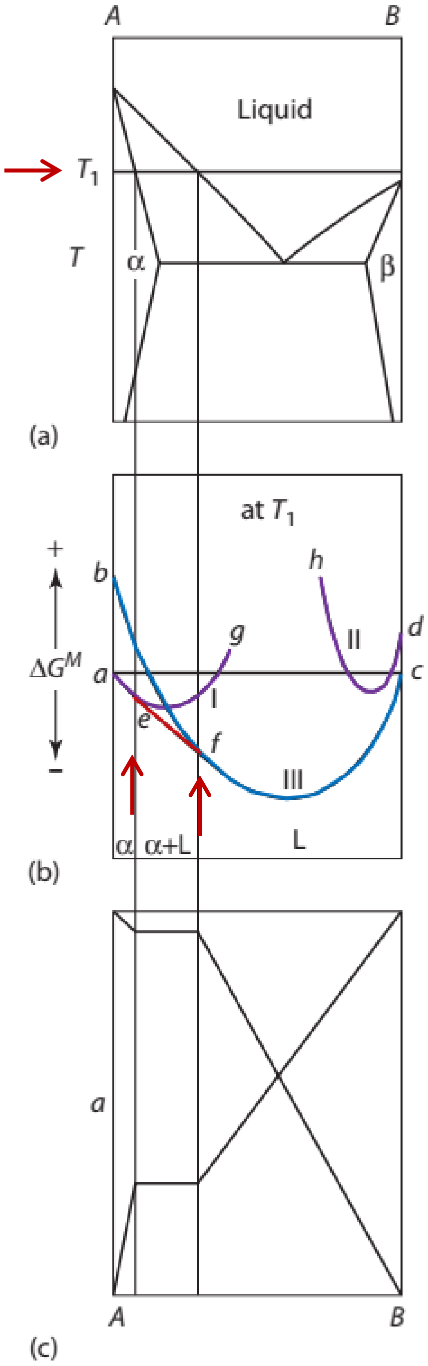
Figure 10.15 the effect of temperature on the molar Gibbs free energies of mixing and the activities of the components of the system *A–B*.

10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity



At T_1 , $T_{m(A)} > T_1 > T_{m(B)}$

- Point ***a***: molar $\Delta G^M = 0$ of pure **Solid *A*** (as the standard state) .
- Point ***c***: molar $\Delta G^M = 0$ of pure **Liquid *B*** (as the standard state) .
- Point ***b***: molar ΔG^M of pure **Liquid *A***.
- Point ***d***: molar ΔG^M of pure **Solid *B***.



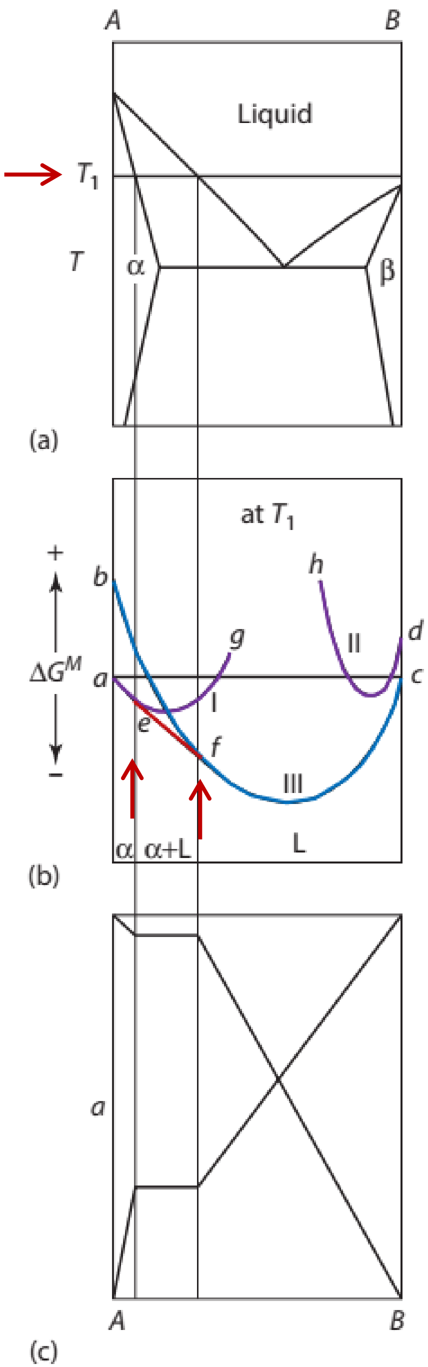
10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity

At T_1 , $T_{m(A)} > T_1 > T_{m(B)}$

- ✓ The curve ***aeg*** (curve **I**) is the ΔG^M of solid *A* and solid *B* to form homogeneous ***α*** solid solutions, which have the same crystal structure as has *A*.

- ✓ This curve intersects the $X_B = 1$ axis at the molar Gibbs free energy which solid *B* would have if it had the same crystal structure as has *A*.

10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity



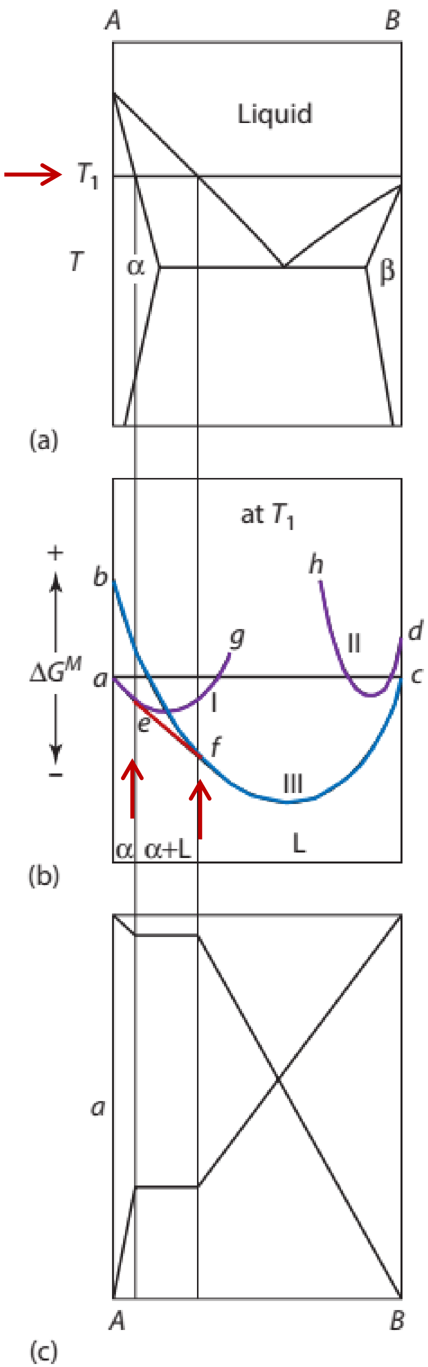
At T_1

✓ the **curve *dh*** (curve **II**) represents the ΔG^M of solid B and solid A to form homogeneous β solid solutions which have the same crystal structure as has B.

✓ This curve intersects the $X_A = 1$ axis at the molar Gibbs free energy which A would have if it had the same crystal structure as B.

✓ The **curve *bfc*** (curve **III**) represents the molar ΔG^M of liquid A and liquid B to form a homogeneous liquid solution.

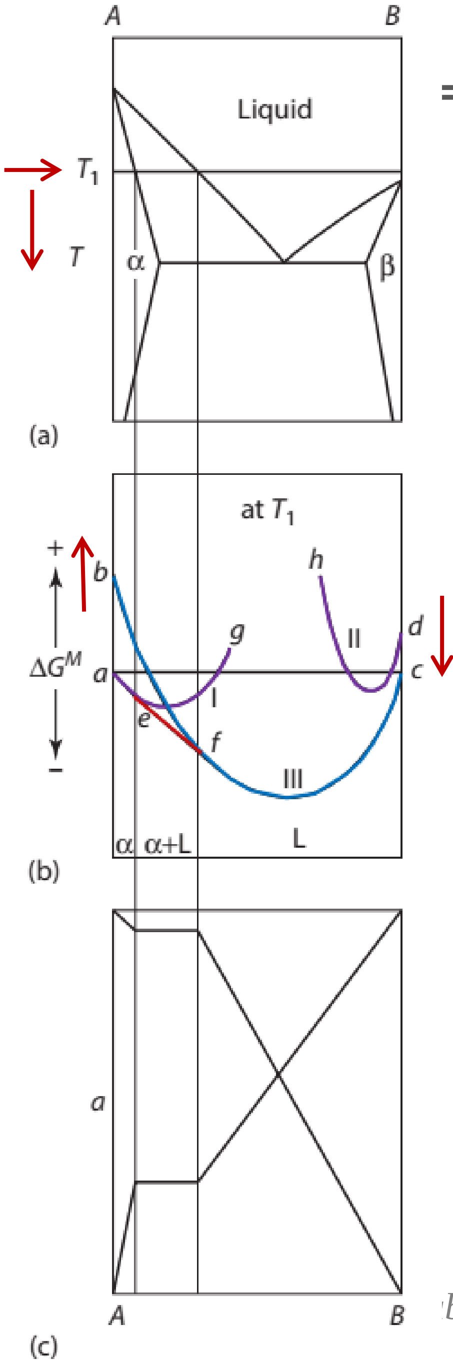
10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity



At T_1

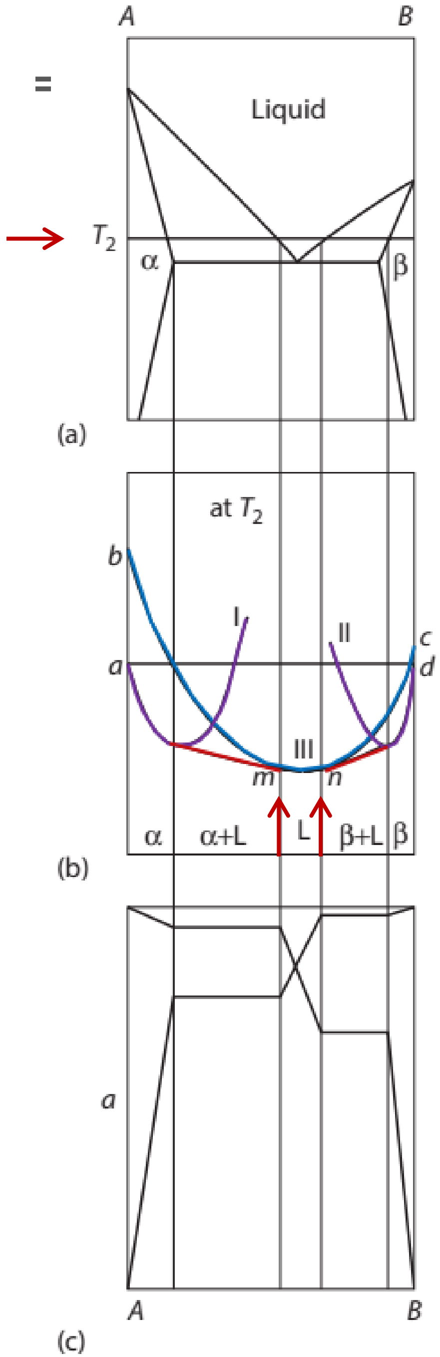
- ✓ Since **curve II** lies everywhere above **curve III**, solid ***β*** solutions are not stable at the temperature T_1.
- ✓ The common tangent to the **curves I** and **III** identifies the **α** solidus composition at the temperature T_1 as ***e*** and the liquidus composition as ***f***.

10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity



At T_1

- ✓ Figure 10.15c shows the ***a–X_B*** relationships of the components at the temperature ***T_1***. These relationships are drawn in accordance with the **assumption** that the liquid solutions exhibit Raoultian ideality and the solid solutions show positive deviations from Raoult's law.
- ✓ As the temperature decreases **below *T_1***, the length of ***ab*** increases and the length of ***cd*** decreases until, at ***$T = T_{m(B)}$***, the points ***c*** and ***d*** **coincide** at $\Delta G^M = 0$.
- ➢ Temperature decreases: point ***d*** approaches ***c***



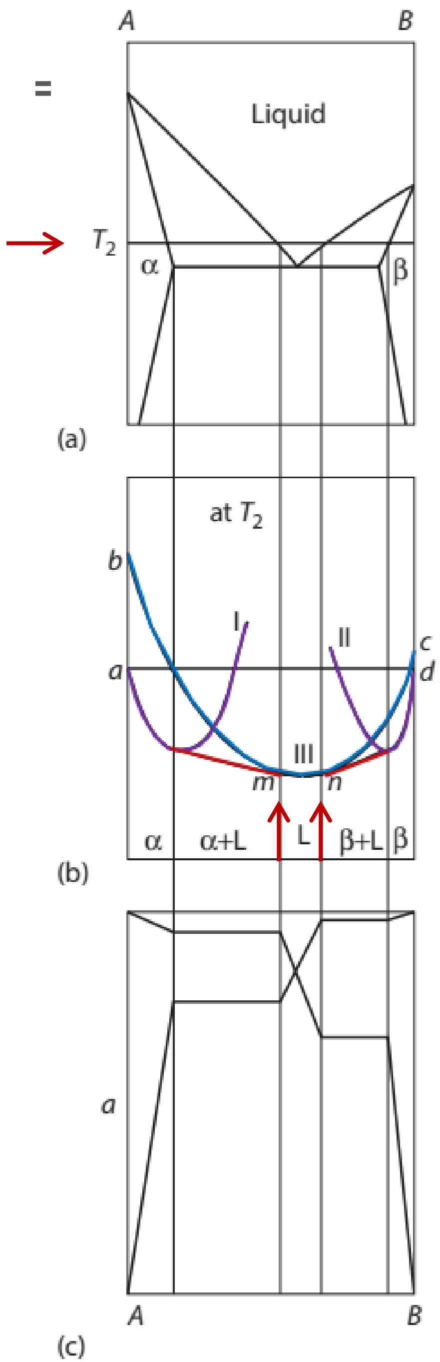
At T_2 , $T_{m(B)} > T_2 > T_E$

✓ At $T_2 < T_{m(B)}$ the point ***c*** (liquid B) lies above ***d*** in Figure 10.16b, and

- Point ***a***: pure **Solid *A*** (as the standard state) .
- Point ***b***: pure **Liquid *A***.
- Point ***c***: pure **Liquid *B***.
- Point ***d***: pure **Solid *B*** (as the standard state) .

➢ The a–X_B curves at T_2 are shown in Figure 10.16c, in which the solid is the standard state for both components.

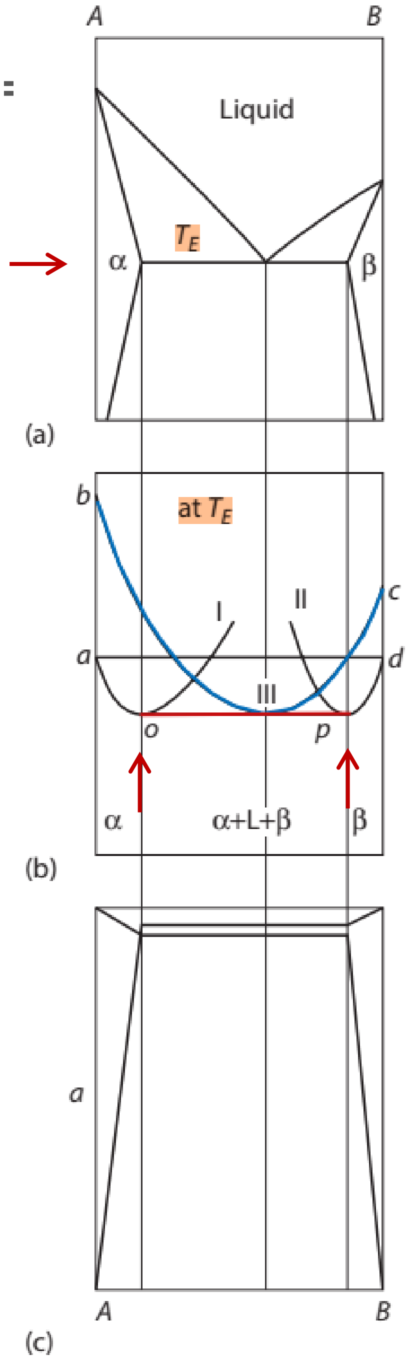
10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity



At T_2 , $T_{m(B)} > T_2 > T_E$

- ✓ At $T_2 < T_{m(B)}$
- ✓ since curve **II** lies partially below curve **III**, **two common tangents** can be drawn:
- ✓ one to the curves I and III, which defines the compositions of the solidus ***α*** and its conjugate liquidus, and
- ✓ one to the curves II and III, which defines the compositions of the solidus ***β*** and its conjugate liquidus.

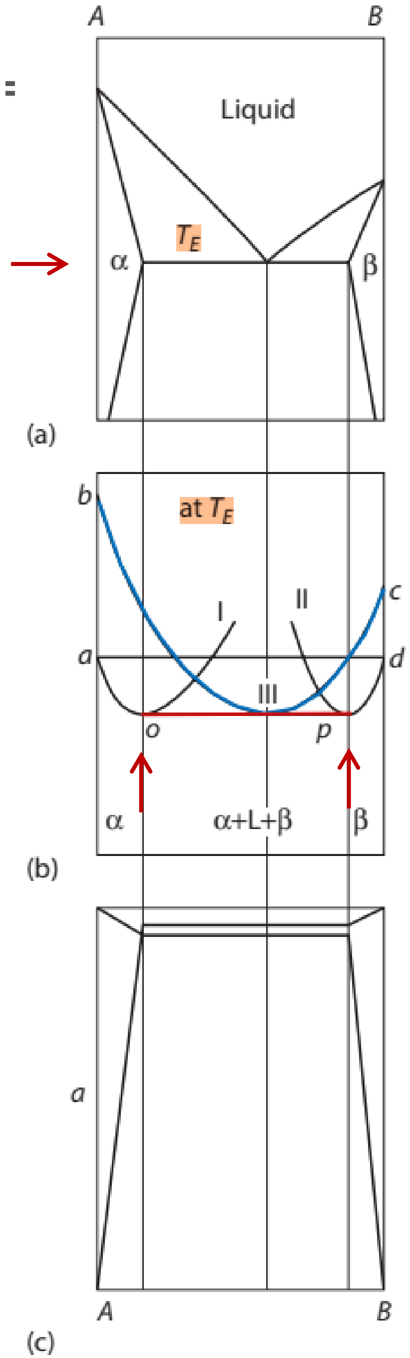
10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity



At $\boldsymbol{T_E}$: *eutectic T*

- ✓ With ***T*** further decrease,
- ✓ the two liquidus compositions ***m*** and ***n*** in Figure 10.16b **approach one another** and, at the unique temperature $\boldsymbol{T_E}$ (the eutectic *T*), the **two common tangents** merge to form the "**triple**" common tangent to the three curves shown in Figure 10.17b.

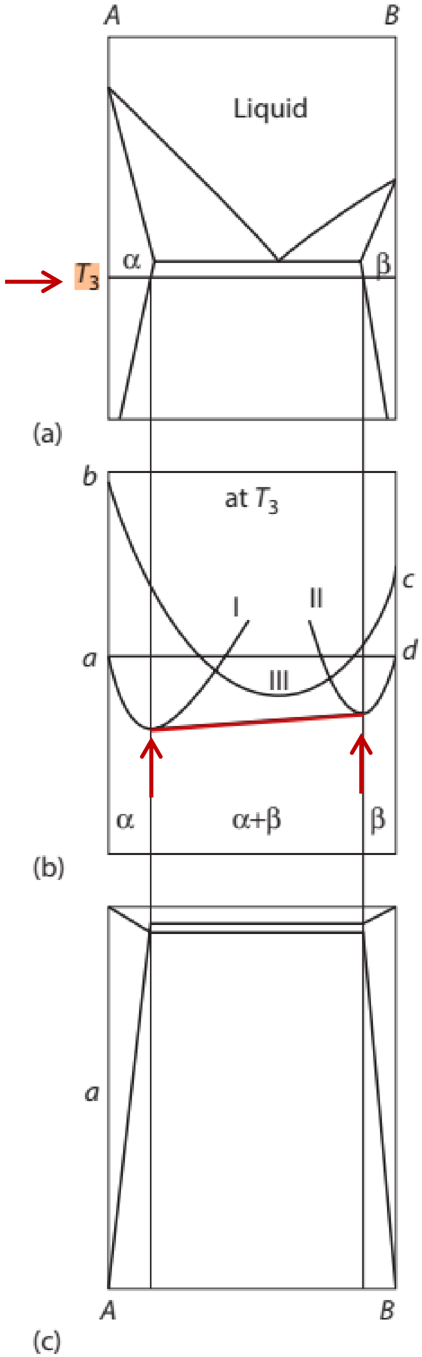
10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity



At $\boldsymbol{T_E}$: *eutectic T*

- ✓ At compositions between ***o*** and ***p***, a **doubly** saturated eutectic **liquid coexists** in equilibrium with ***α*** and ***β*** solid solutions.
- ✓ From the Gibbs phase rule, this **three-phase equilibrium** has **one** degree of freedom, which is used to specify the pressure of the system.
- ✓ Thus, at the specified pressure, the three-phase equilibrium is invariant (the compositions of the three phases are fixed and the ***T*** is fixed).

10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity

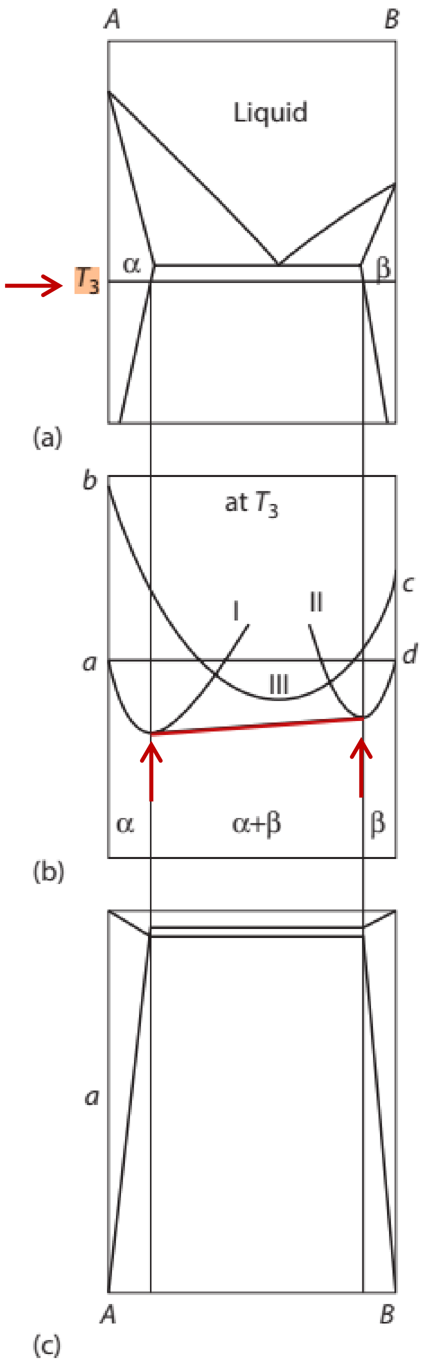


$T_3 < T_E$

- At $T_3 < T_E$,
- **curve** III lies above the common tangent to **curves I** and **II** (Figure 10.18b) and thus,
- the liquid phase is not stable.
- The corresponding activity–composition relationships are shown, respectively, in Figure 10.18 c.

Figure 10.18
the effect of temperature on the molar Gibbs free energies of mixing and the activities of the components of the system *A*–*B*.

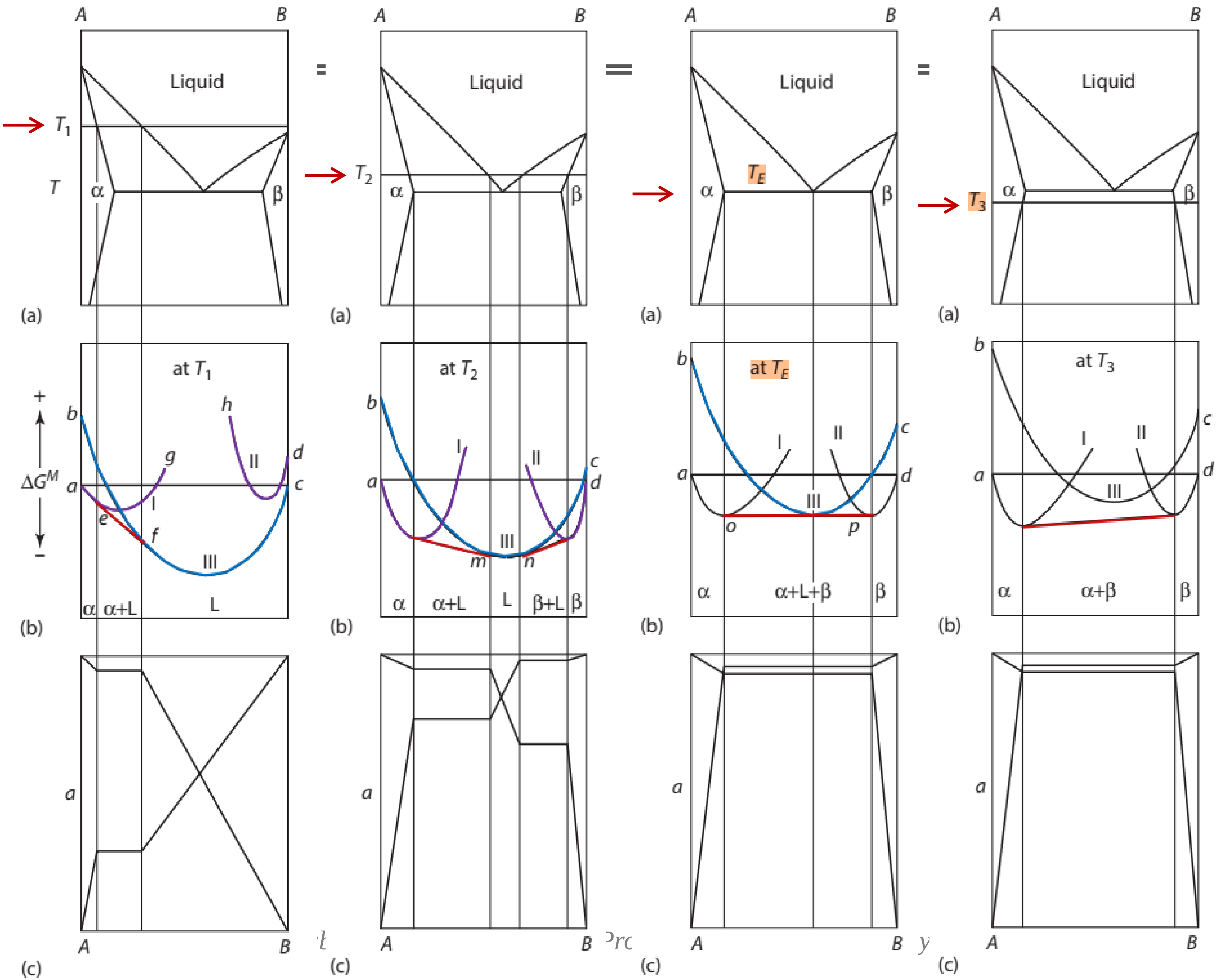
10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity



$T_3 < T_E$

- ✓ If the ranges of **solid solubility** in the ***α*** and ***β*** phases are **immeasurably small**, then, as a reasonable approximation,
- ✓ it can be said that *A* and *B* are **insoluble** in one another in the solid state.

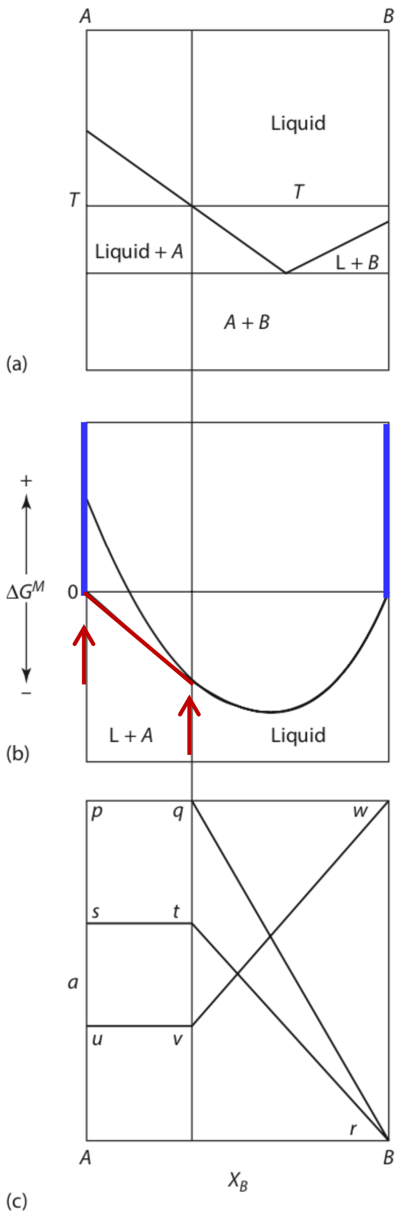
Figure 10.18
the effect of temperature on the molar Gibbs free energies of mixing and the activities of the components of the system *A*–*B*.



$T \downarrow$

$T_1 \downarrow T_2 \downarrow T_E \downarrow T_3$

10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity



✓ The phase diagram for such a system (*A* and *B* are **insoluble**) is shown in Figure 10.19a.

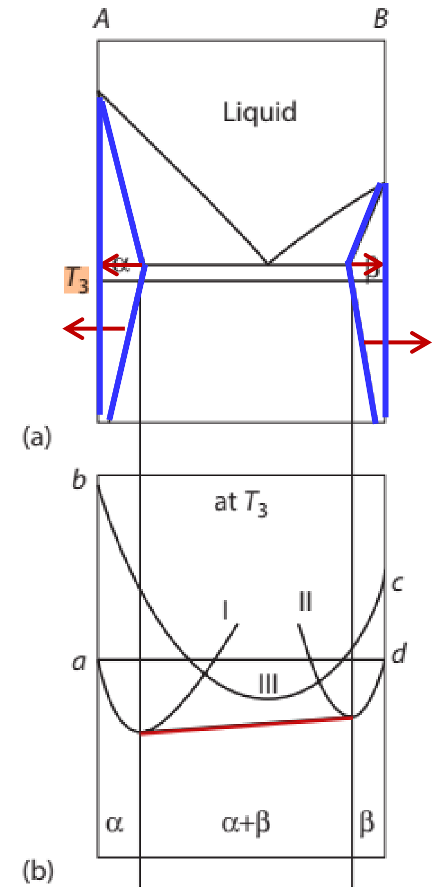
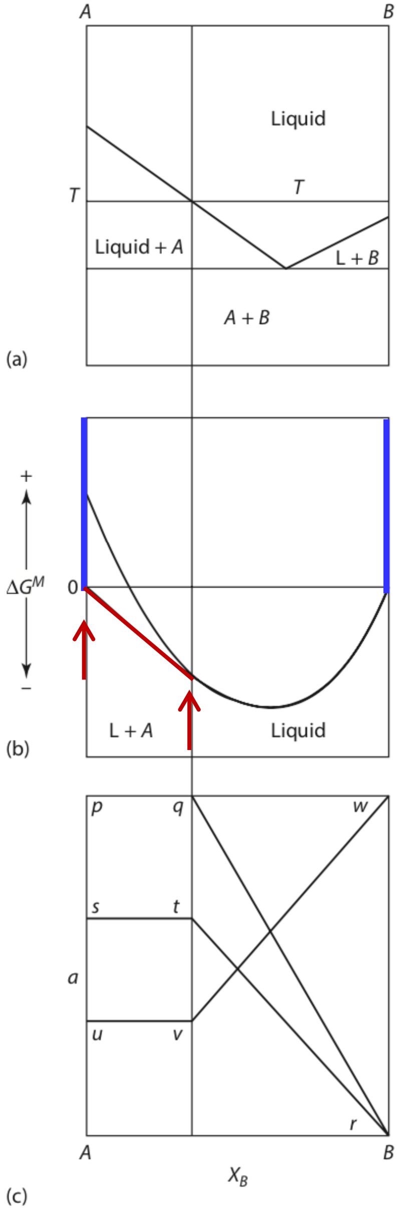


Figure 10.19
the molar Gibbs free energy of mixing and the activities in a binary eutectic system that exhibits complete liquid miscibility and virtually complete solid immiscibility.

10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity



- ✓ Since the range of solid solubility in Figure 10.19a is so small that it may be neglected on the scale of Figure 10.19a, then the molar Gibbs free energy of mixing curves for the formation of ***α*** and ***β*** (**curves I** and **II** in Figures 10.15 through 10.18) are also compressed toward the ***X_A* = 1** and ***X_B* = 1** axes, respectively.
- ✓ On the scale of Figures 10.15 through 10.18, they coincide with the **vertical axes**.

- ✓ The sequence in Figure 10.20 shows how, as the **solubility of *B*** in **α decreases**, the Gibbs free energy curve for **α** is compressed against the $\boldsymbol{X_A = 1}$ axis.

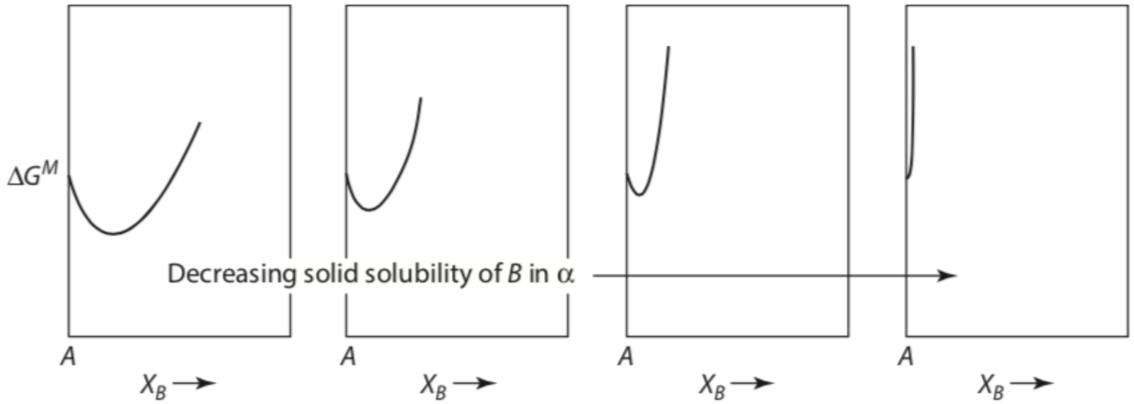
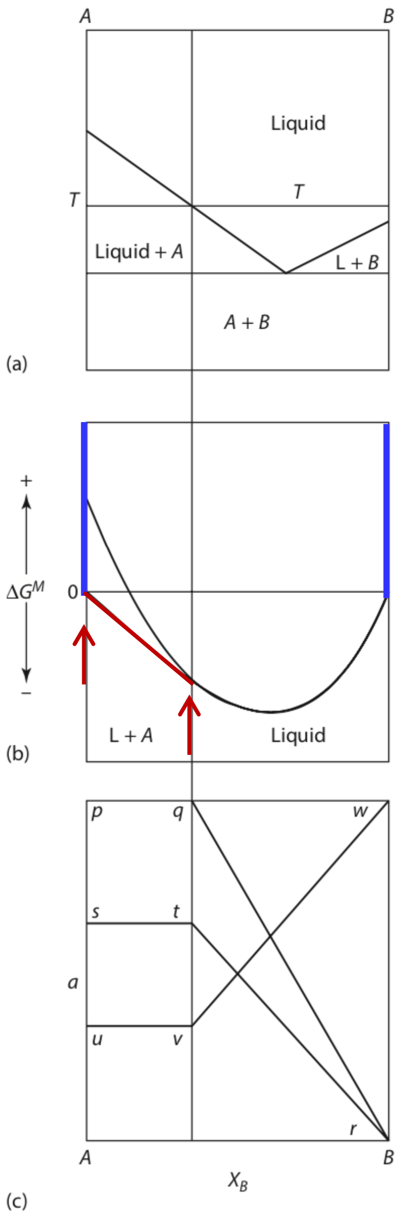


Figure 10.20

the effect of decreasing solid solubility on the molar Gibbs free energy of mixing curve.



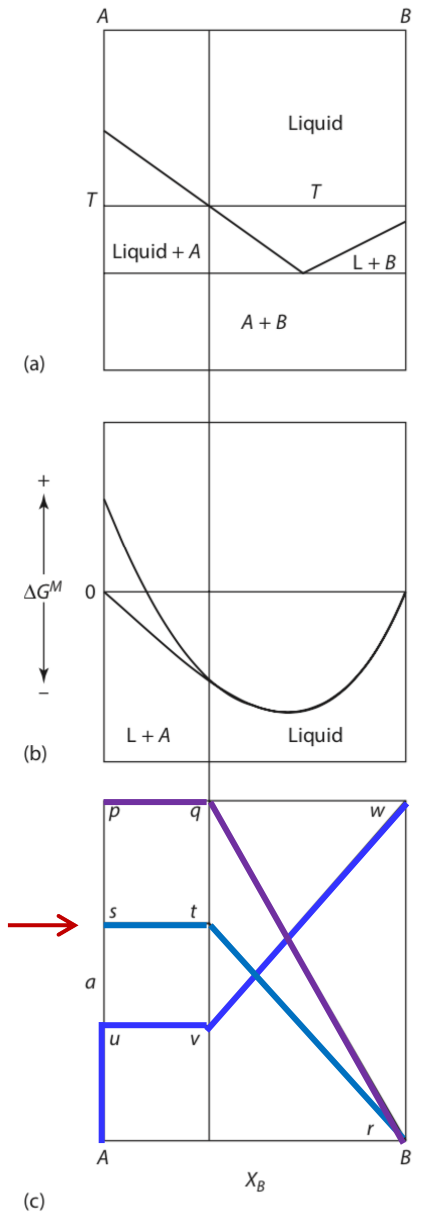
- ✓ The Gibbs free energy of formation of the liquid solutions in the system *A*–*B* at the temperature ***T*** is shown in Figure 10.19b.

- ✓ The **common tangent** to the **α** solid solution and liquid solution curves is reduced to a tangent drawn from the point on the **X_A = 1 axis** which represents pure solid *A* to the liquid solutions curve.

Figure 10.19

the molar Gibbs free energy of mixing and the activities in a binary eutectic system that exhibits complete liquid miscibility and virtually complete solid immiscibility.

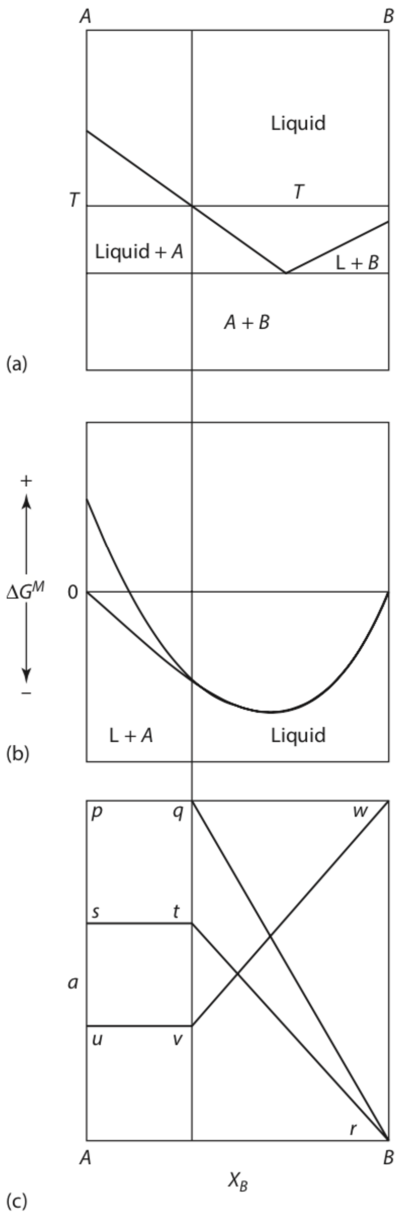
10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity



- ✓ In Figure 10.19c (***a–X***$_B$ relations)
- ✓ ***pqr*** is the ***a***$_A$ with respect to pure solid *A* at ***p***,
- ✓ ***s*** is the activity of pure liquid *A* with respect to solid *A* at ***p***,
- ✓ ***str*** is the ***a***$_A$ with respect to liquid *A* having unit activity at ***s***, and
- ✓ ***Auvw*** is the ***a***$_B$ with respect to liquid *B* having unit activity at ***w***.

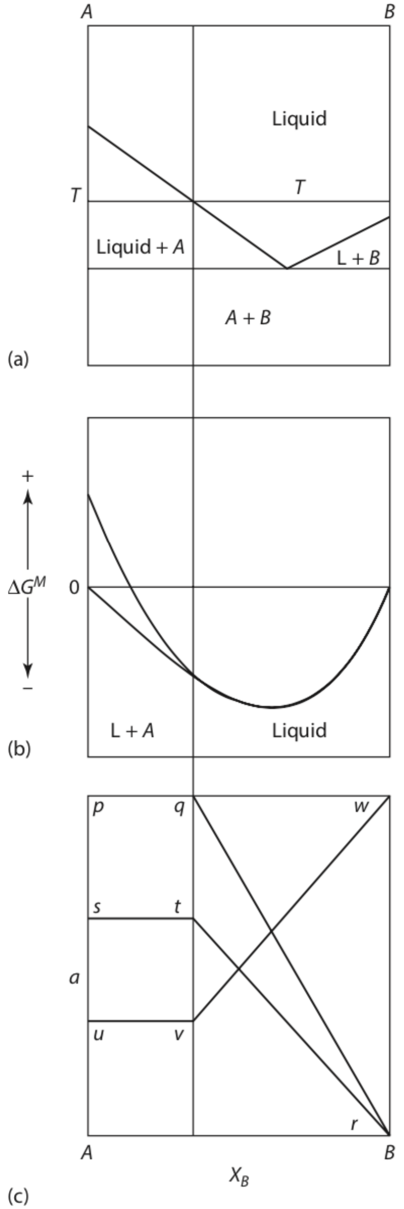
Figure 10.19
the molar Gibbs free energy of mixing and the activities in a binary eutectic system that exhibits complete liquid miscibility and virtually complete solid immiscibility.

10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity



✓ In a binary system which exhibits complete miscibility in the liquid state and virtually complete immiscibility in the solid state (e.g., Figure 10.19a), the variations of the activities of the components of the liquid solutions can be obtained from consideration of the liquidus curves.

10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity



- ✓ At any temperature *T* (Figure 10.19a), the system with a composition between pure *A* and the liquidus composition exists as virtually pure solid *A* in equilibrium with a liquid solution of the liquidus composition. Thus, at *T*,

$$G^0_{A(s)} = \bar{G}^A_{(l)} = G^0_{A(l)} + RT\ln a_A$$

- ✓ in which a_A is with respect to liquid *A* as the standard state. Thus,

$$\Delta G^0_{m(A)} = -RT\ln a_A$$

- ✓ or, if the liquid solutions are Raoultian,

$$\Delta G^0_{m(A)} = -RT\ln X_A$$

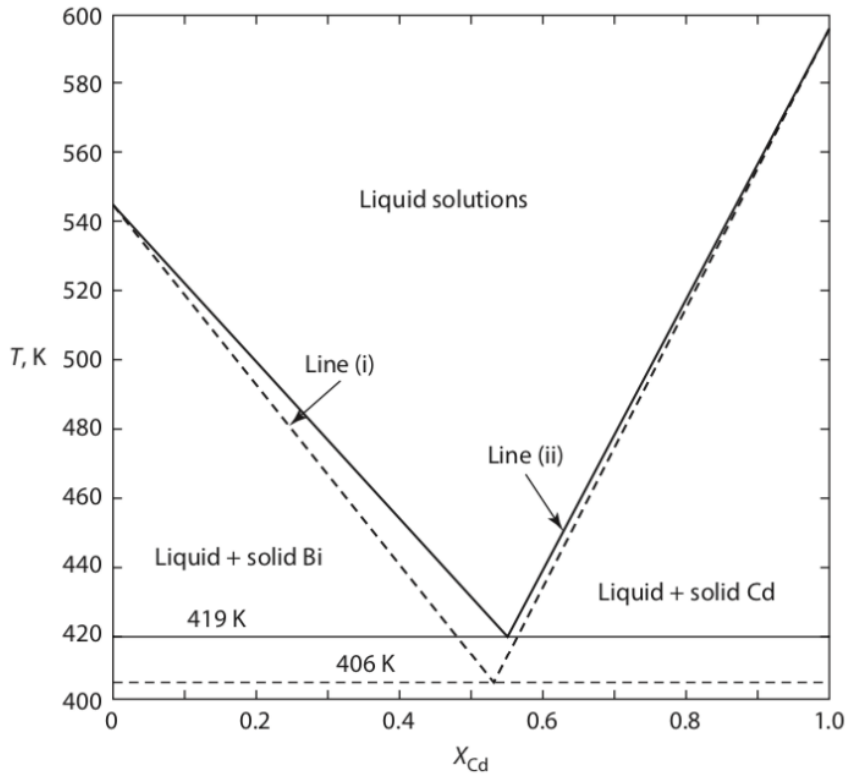


Figure 10.21

the phase diagram for the system Bi–Cd. the full lines are the measured liquidus lines, and the broken lines are calculated assuming no solid solution and ideal mixing in the liquid solutions.

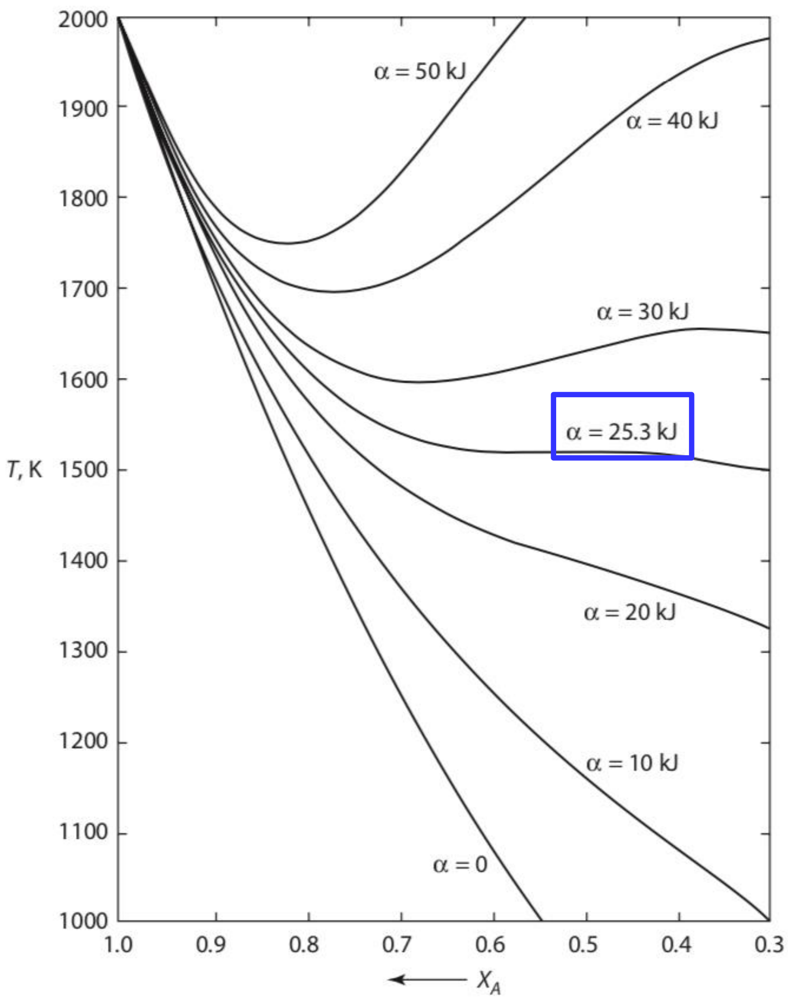
10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity

- ✓ Figure 10.21 shows the binary phase diagram for the Bi–Cd system. This system shows the limited solubility of Bi in Cd and virtually zero solubility of Cd in Bi..
- ✓ It is of interest to examine what happens to the liquidus line as the magnitude of the positive deviation from Raoultian behavior in the liquids increases—that is, as G^{XS} becomes increasingly positive.
- ✓ Assuming regular solution behavior, Equation 10.11, written in the form

$$\Delta G^0_{m(A)} = -RT\,\underline{\ln a_A} \longrightarrow -\Delta G^0_{m(A)} = RT\ln X_A + RT\,\underline{\ln \gamma_A}$$

$$\downarrow$$

$$-\Delta G^0_{m(A)} = RT\ln X_A + \alpha\left(1 - X_A\right)^2$$

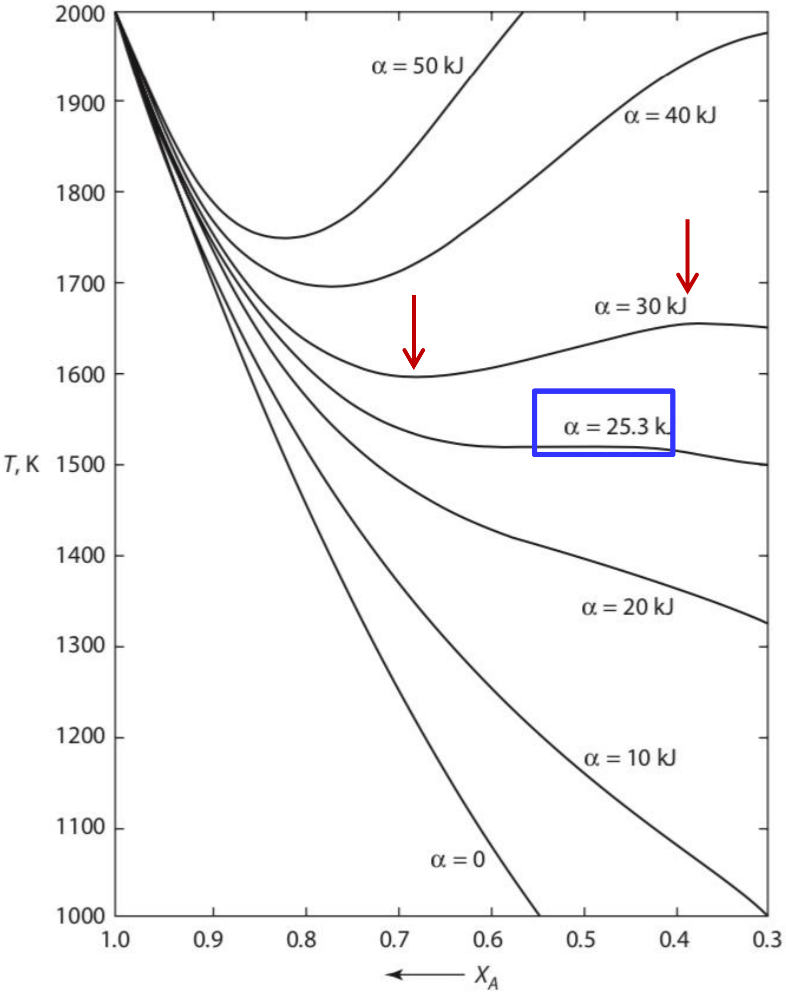


- ✓ Consider a **hypothetical** system *A–B* in which $\Delta H^o_{m(A)}$ = 10 kJ at $T_{m(A)}$= 2000 K.
- ✓ Thus, for this system,

$$-10,000 + 5T = RT \ln X_A + \alpha \left(1 - X_A\right)^2$$

- ✓ where X_A is the composition of the *A* liquidus at *T*. The *A* liquidus lines, drawn for ***α*** = 0~50 kJ, are shown in Figure 10.22.

Figure 10.22
Calculated liquidus lines assuming regular solution behavior in the liquid solutions and no solid solubility.

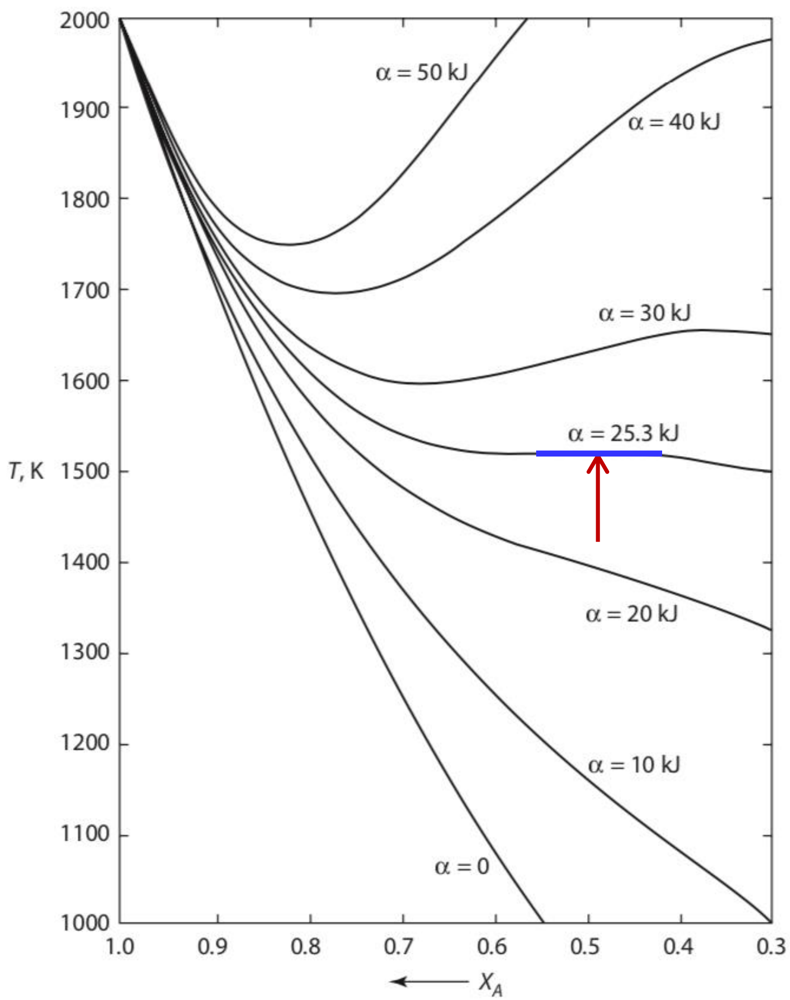


- ✓ As ***α*** exceeds some **critical value** (25.3 kJ), the form of the liquidus line changes from a monotonic decrease in liquidus temperature with decreasing X_A to a form which contains a **maximum** and a **minimum** (e.g., the liquidus for ***α* = 30** kJ).

Figure 10.22
Calculated liquidus lines assuming regular solution behavior in the liquid solutions and no solid solubility.

10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity

✓ At the critical value of ***α***, the **max**imum and **min**imum **coincide** at X_A = 0.5 to produce a horizontal inflection in the liquidus curve.



✓ It is apparent that, when ***α*** exceeds the critical value, isothermal tie-lines **cannot be drawn** between pure solid *A* and all points on the liquidus lines, which, necessarily, means that the calculated liquidus lines are **impossible**.

Figure 10.22
Calculated liquidus lines assuming regular solution behavior in the liquid solutions and no solid solubility.

10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity

- ✓ From Equation 10.6:

$$\ln a_A = \frac{-\Delta H^0_{m(A)}}{RT} + \frac{\Delta H^0_{m(A)}}{RT_{m(A)}}$$

- ✓ Thus,

$$d\ln a_A = \frac{da_A}{a_A} = \frac{\Delta H^0_{m(A)}}{RT^2}dT \qquad \text{or} \qquad \frac{dT}{dX_A} = \frac{RT^2}{\Delta H^0_{m(A)}a_A}\underline{\frac{da_A}{dX_A}}$$

- ✓ and also,

$$\frac{d^2T}{dX_A^2} = \frac{2RT}{\Delta H^0_{m(A)}a_A}\underline{\frac{da_A}{dX_A}}\frac{dT}{dX_A} - \frac{RT^2}{\Delta H^0_{m(A)}a_A^2}\underline{\left(\frac{da_A}{dX_A}\right)^2} + \frac{RT^2}{\Delta H^0_{m(A)}a_A}\underline{\frac{d^2a_A}{dX_A^2}}$$

$da_A/dX_A = d^2a_A/dX^2_A$ **=0** at the state of imminent immiscibility.

- ✓ Thus,

$dT/dX_A = d^2T/dX^2_A$ **= 0** at the state of imminent immiscibility.

- In Figure 10.22, $\alpha_{cr,r}$**=25.3** kJ, and the horizontal inflection in the critical liquidus curve occurs at X_A =0.5, T=1413 K.

- Thus,

$$\frac{\alpha_{cr}}{RT_{cr}} = \frac{25{,}390}{8.3144 \times 1413} = 2$$

- which is in accord with Equation 10.8.

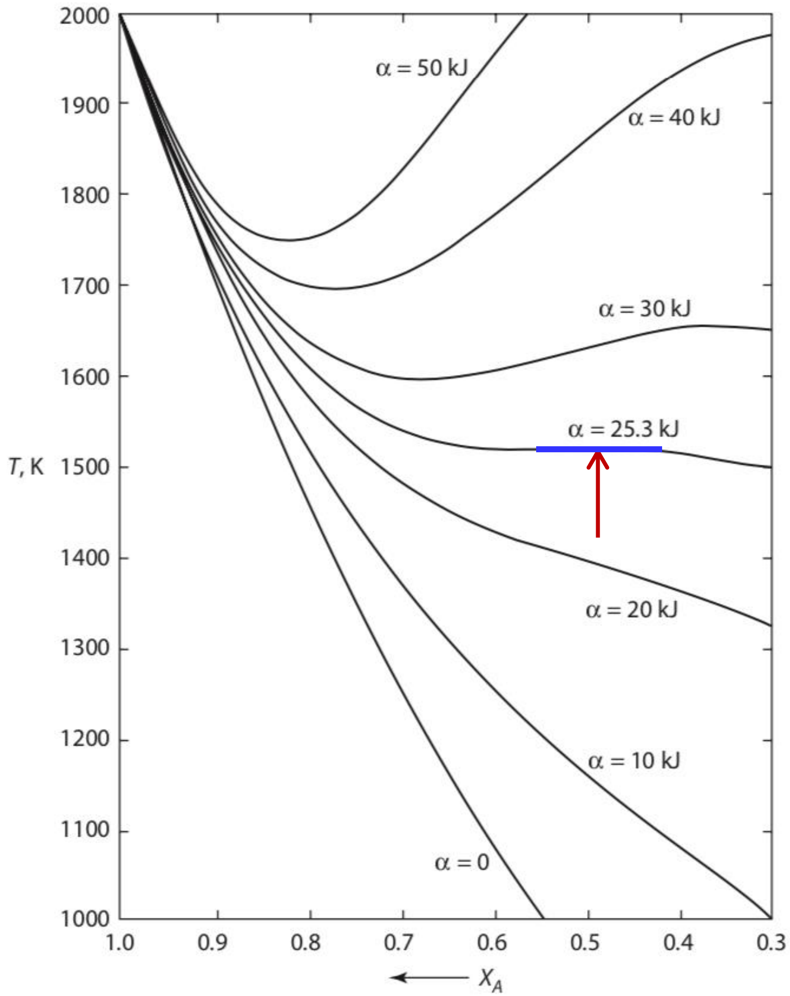
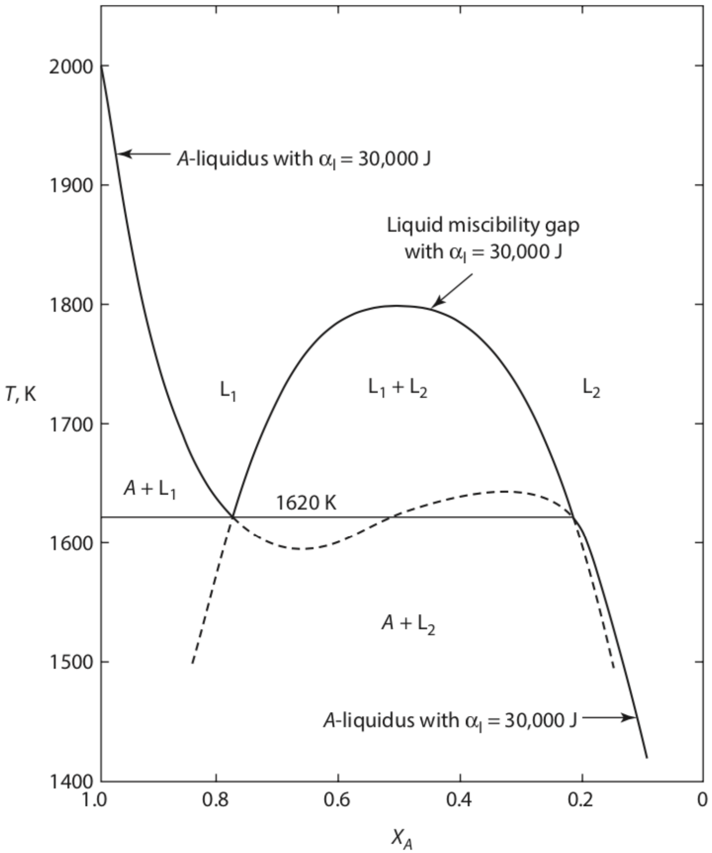


Figure 10.22
Calculated liquidus lines assuming regular solution behavior in the liquid solutions and no solid solubility.

10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity



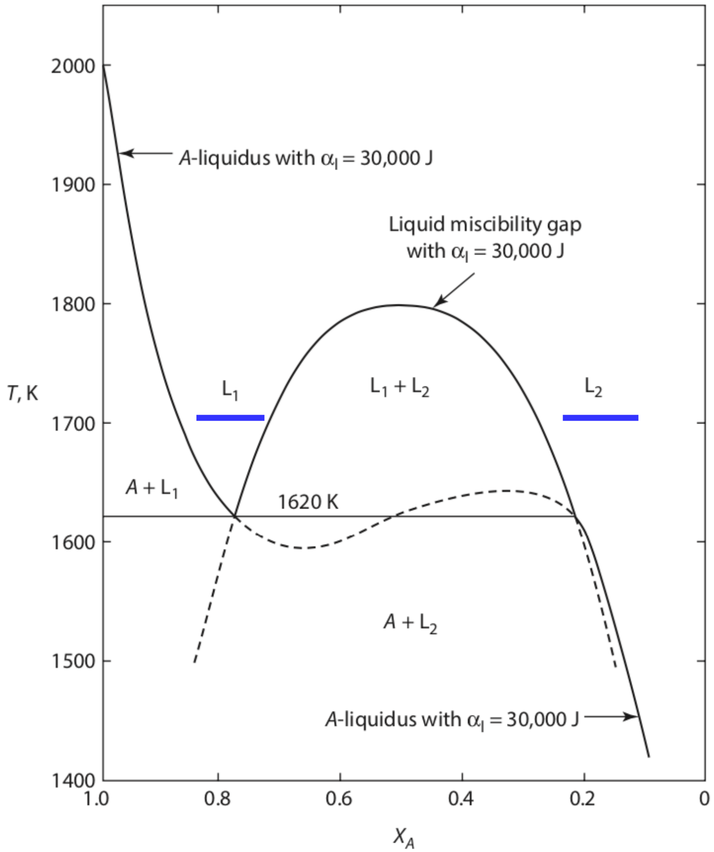
- ✓ The phase equilibria generated when $\alpha_{cr} > \alpha_{cr,\ r}$ are shown in Figure 10.23 which shows the immiscibility in regular liquid solutions with α_{cr} = 30 kJ and the A-liquidus for α = 30 kJ shown in Figure 10.22.

Figure 10.23

the monotectic equilibrium in a binary system in which the liquid solutions exhibit regular solution behavior with α_l = 30,000 J.

10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity

- ✓ The liquid immiscibility curve and the A-liquidus curve intersect at 1620 K to produce a three-phase monotectic equilibrium between *A* and liquidus ***L_1*** and ***L_2***.



- ✓ The liquid immiscibility curve is metastable at temperatures less than 1620 K, and
- ✓ the calculated A-liquidus is physically impossible between the compositions of ***L_1*** and ***L_2*** at 1620 K.

Figure 10.23

the monotectic equilibrium in a binary system in which the liquid solutions exhibit regular solution behavior with α_l = 30,000 J.

Contents

10.8 the Phase Diagrams of Binary Systems That Exhibit Regular Solution Behavior in The Liquid And Solid States

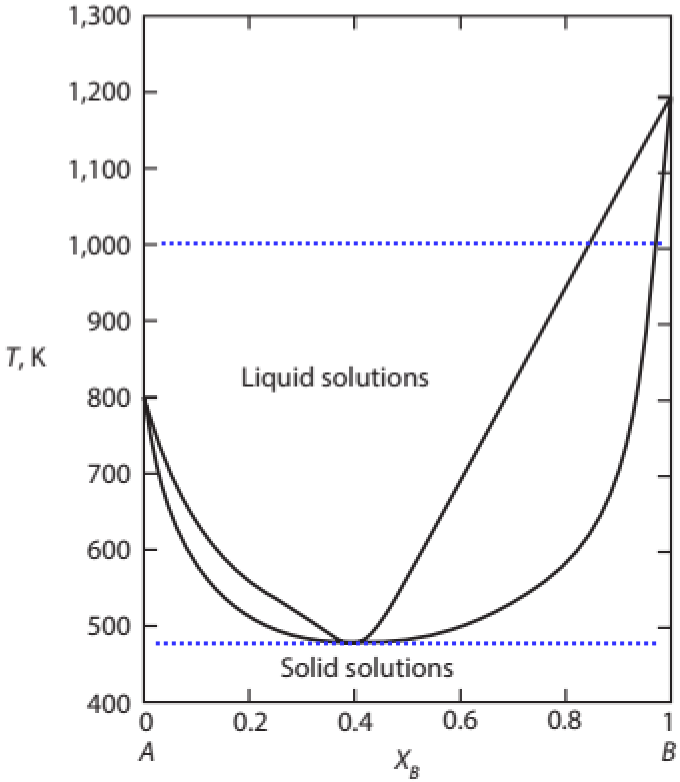
- ✓ Consider the binary system *A*–*B* which forms regular liquid solutions and regular solid solutions.
- ✓ $T_{m,(A)}$ = 800 K and $T_{m,(B)}$ = 1200 K
- ✓ the molar Gibbs free energies of melting in ***J*** are

$$\Delta G_{m,(A)} = 8000 - 10T$$

$$\Delta G_{m,(B)} = 12,000 - 10T$$

- ✓ Consider the system in which α_l **= –20,000** J in the liquid solutions and α_s **= 0** in the solid solutions.

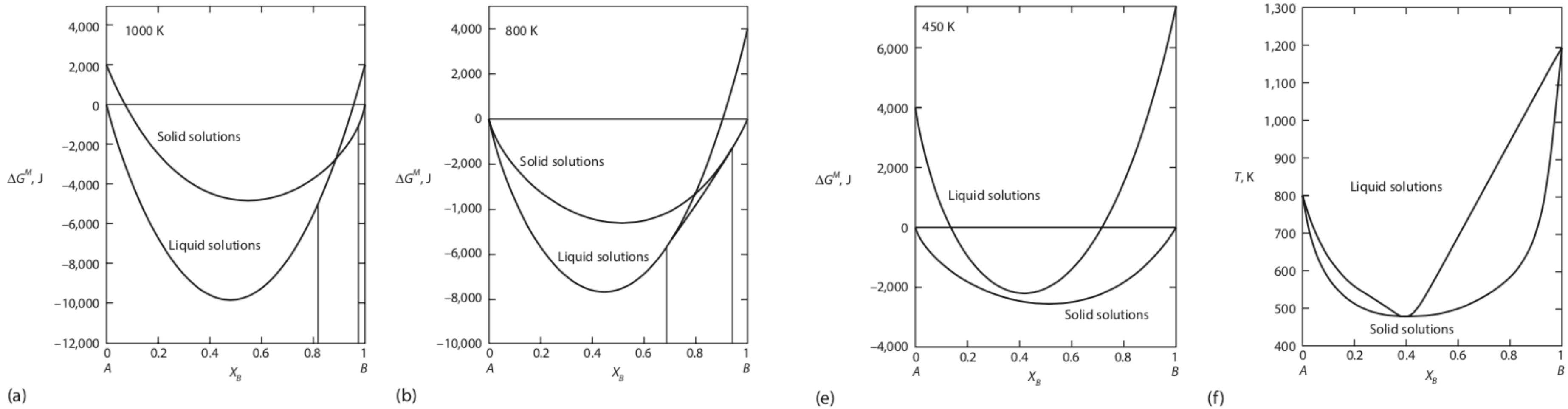
10.8 the Phase Diagrams of Binary Systems That Exhibit Regular Solution Behavior in The Liquid And Solid States

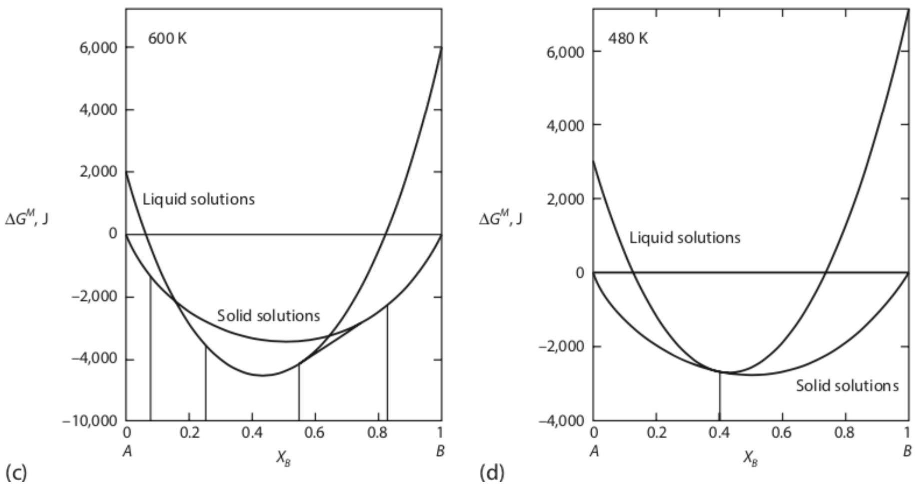


T =1000, 800, 600, **480**, 450 K

Figure 10.24 (**f**) the phase diagram for a binary system which forms regular solid solutions (***α_s* = 0**) and regular liquid solutions (***α_l* = –20 kJ**).

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States





T =1000, 800, 600, **480**, 450 K

Figure 10.24 (a~e) the molar ΔG^M curves **~** ***T***, and (**f**) the phase diagram for a binary system which forms regular solid solutions (**α_s = 0**) and regular liquid solutions (**α_l = -20 kJ**).

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

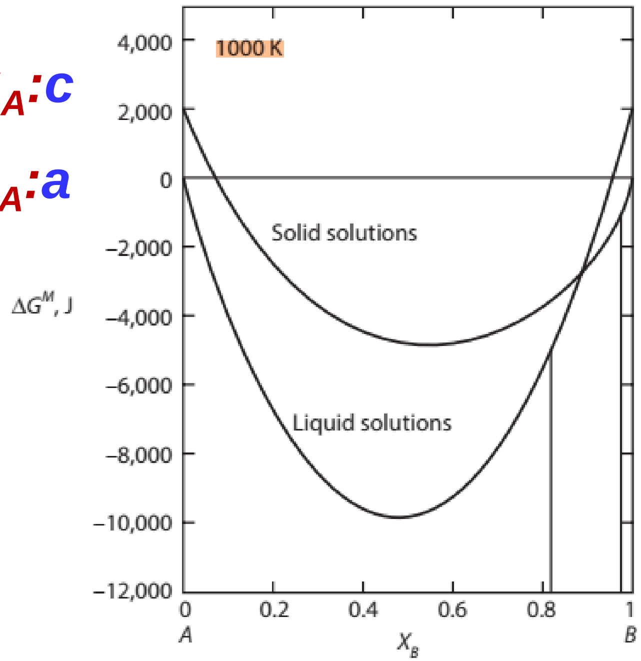
T=1000 K

✓ The **ΔG^M** curves at 1000 K are shown in Figure 10.24a.

✓ Since $T_{m,(A)}$ < *1000 K* < $T_{m,(B)}$, the liquid is chosen as the standard state for *A*, and the solid is chosen as the standard state for *B*.

S_A:c

L_A:a



(a)

d:L_B

b:S_B

Point ***a***: pure Liquid *A* (standard state).

Point ***b***: pure Solid *B* (standard state).

Point ***c***: pure Liquid *B*.

Point ***b***: pure Liquid *A*.

Figure 10.24 (a~e) the molar ΔG^M curves ~ T for a binary system which forms regular solid solutions (α_s = 0 and α_l = –20 kJ). (a) T=1000 K

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

T=1000 K

- With reference to these standard states,
- the ΔG^M are

$$\Delta G_l^M = X_B \Delta G_{m,(B)}^{\circ} + RT(X_A \ln X_A + X_B \ln X_B) + \alpha_l X_A X_B$$

- for the liquid solutions
- and

$$\Delta G_s^M = -X_A \Delta G_{m,(A)}^{\circ} + RT(X_A \ln X_A + X_B \ln X_B) + \alpha_s X_A X_B$$

- for the solid solutions.

- Inserting the numerical data gives

$$\Delta G_l^M = (12,000 - 10T)X_A + 8.3144T(X_A \ln X_A + X_B \ln X_B) - 20,000 X_A X_B$$

- and

$$\Delta G_s^M = -(8000 - 10T)X_A + 8.3144T(X_A \ln X_A + X_B \ln X_B)$$

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

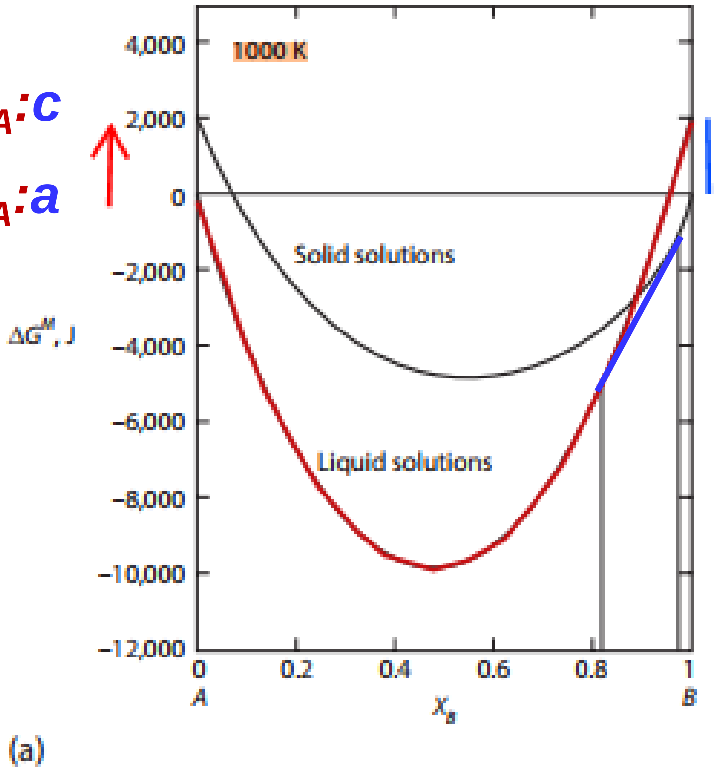
T=1000 K

✓ The common tangent to the curves in Figure 10.24a gives the liquidus composition of X_B = 0.82 and the solidus composition of X_B = 0.97.

✓ Decreasing the temperature causes the Gibbs free energies of the liquids to increase relative to those of the **solids**.

S_A:c

L_A:a



d:L_B

b:S_B

T ↓

L increase towards S

a approaches c ***and then***

away from c

d away from b

Figure 10.24 (a~e) the molar ΔG^M curves ~ T for a binary system which forms regular solid solutions (α_s = 0 and α_l = –20 kJ). (a) T=1000 K

10.8 the Phase Diagrams of Binary Syst
Regular Solution Behavior in the Liquid a

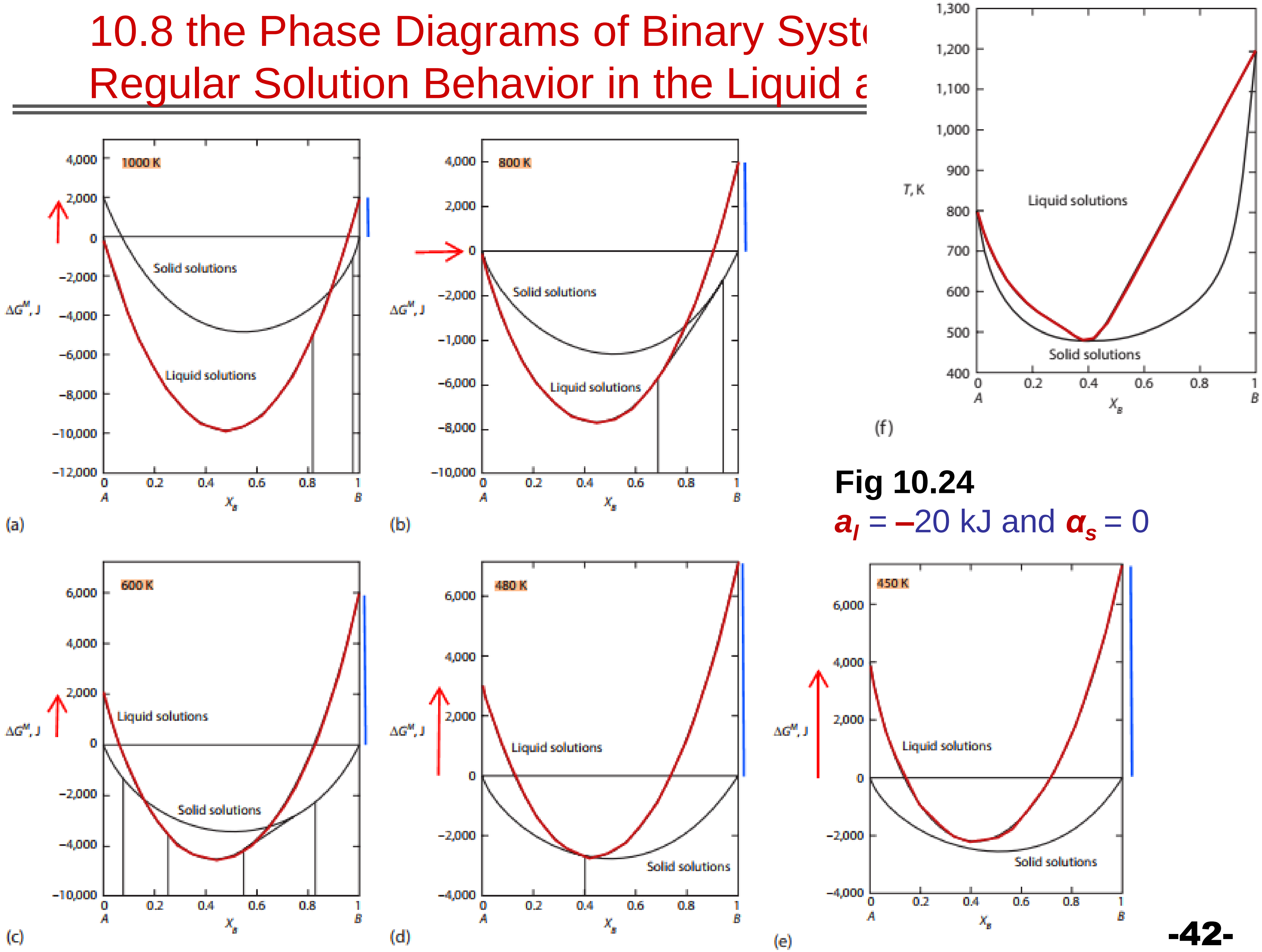


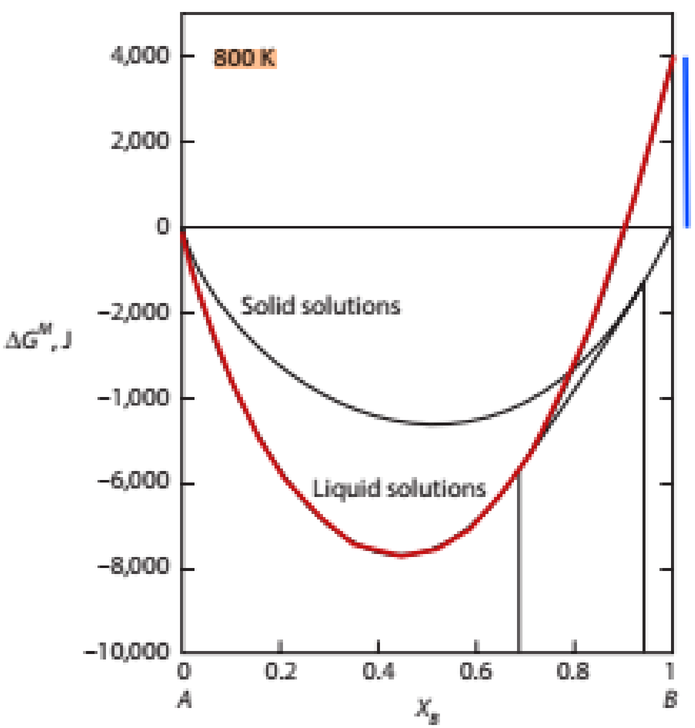
Fig 10.24

a_l = –20 kJ and α_s = 0

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

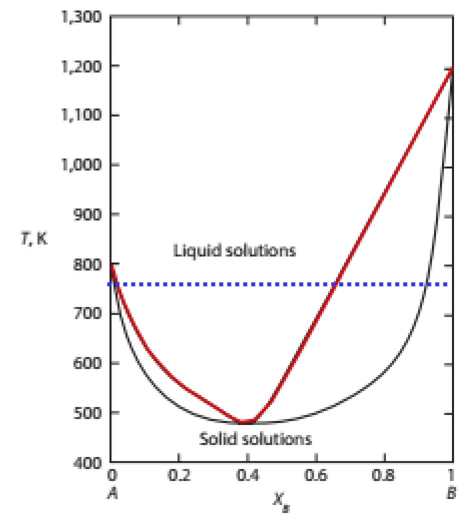
T=800 K

✓ As shown in Figure 10.24b, the Gibbs free energies of pure solid *A* and pure liquid *B* are equal at the melting temperature of *A*, and the common tangent gives liquidus and solidus compositions of X_B = 0.69 and X_B = 0.94, respectively.



(b)

- *T*=1000 K :
- X_B = 0.84 and X_B = 0.97
- *T*= 800 K :
- X_B = 0.69 and X_B = 0.94



(f)

Figure 10.24 (a~e) the molar ΔG^M curves ~ *T* for a binary system which forms regular solid solutions ($\alpha_s = 0$ and $\alpha_l = -20$ kJ). (b) *T*=800 K

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

T=600 K

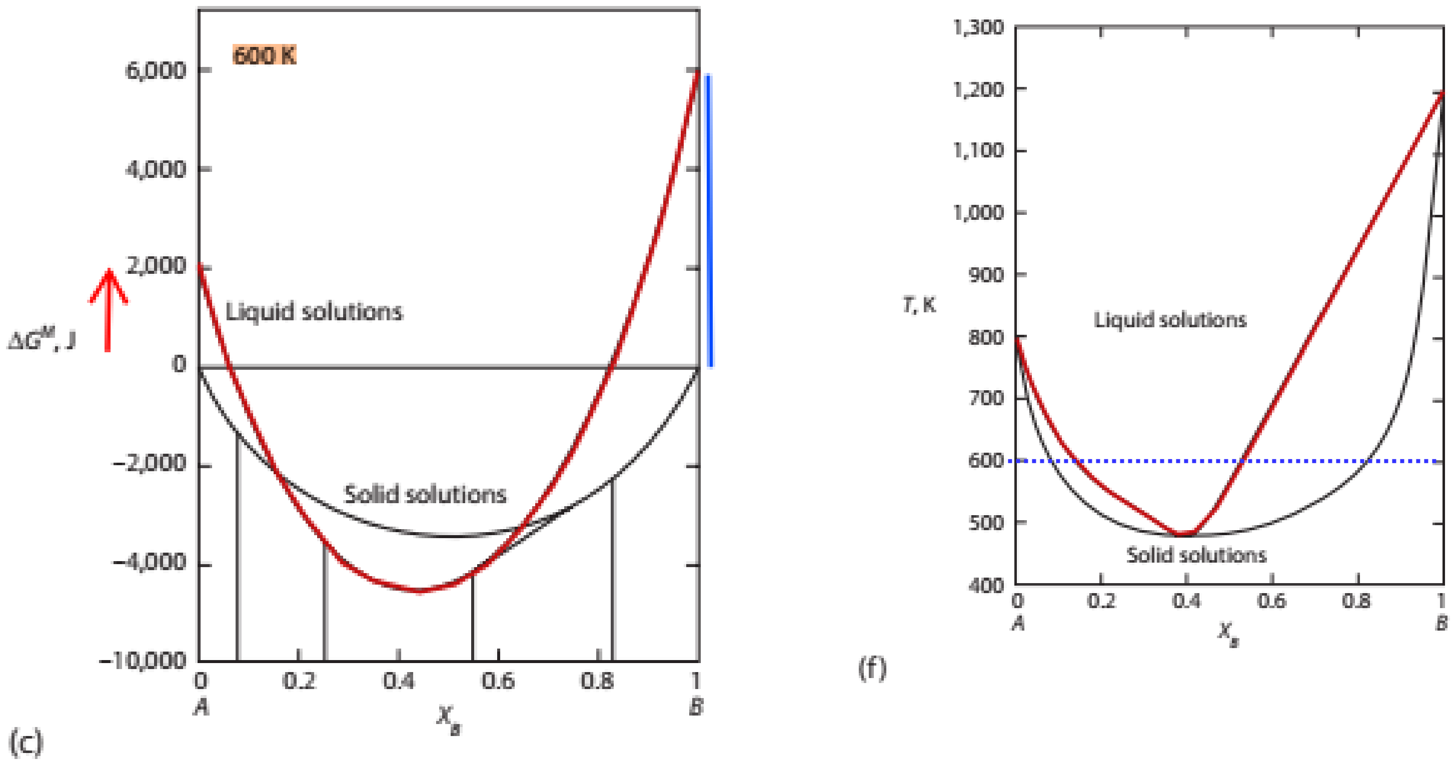
✓ At temperatures lower than $T_{m,(A)}$ and $T_{m,(B)}$, solid is chosen as the standard state for both components and the **ΔG^M** are written as:

$$\Delta G_l^M = (12,000 - 10T)X_A + 8.3144T(X_A \ln X_A + X_B \ln X_B) - 20,000 X_A X_B$$

$$\Delta G_s^M = RT(X_A \ln X_A + X_B \ln X_B)$$

✓ At 600 K, shown in Figure 10.24c, contain **two common tangents**

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States



- ***T*=1000 K : X_B = 0.84 and X_B = 0.97**
- ***T*= 800 K : X_B = 0.69 and X_B = 0.94**
- ***T*= 600 K**
- ***T*= 480 K : X_B = 0.41**

Figure 10.24 (c) *T*=600 K

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

T=480 K

- ✓ At at 480 K, the two double tangents collapse to a **single point** of contact between the curves at $\boldsymbol{X_B = 0.41}$.

T=450 K

- ✓ At temperatures less than 480 K, the curve for the liquids lies **above** that for the solids, and thus, solid solutions are **stable** over the entire range of composition.

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

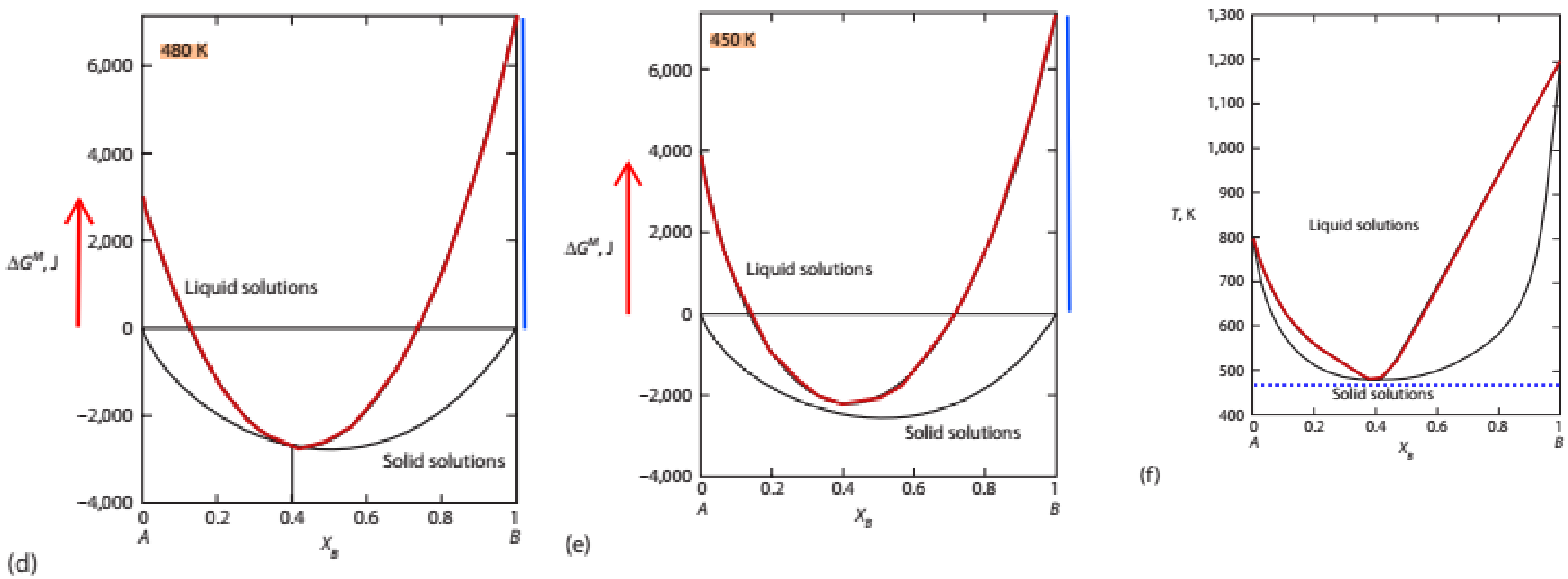


Figure 10.24
(d) *T*=480 K
(e) *T*=450 K

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

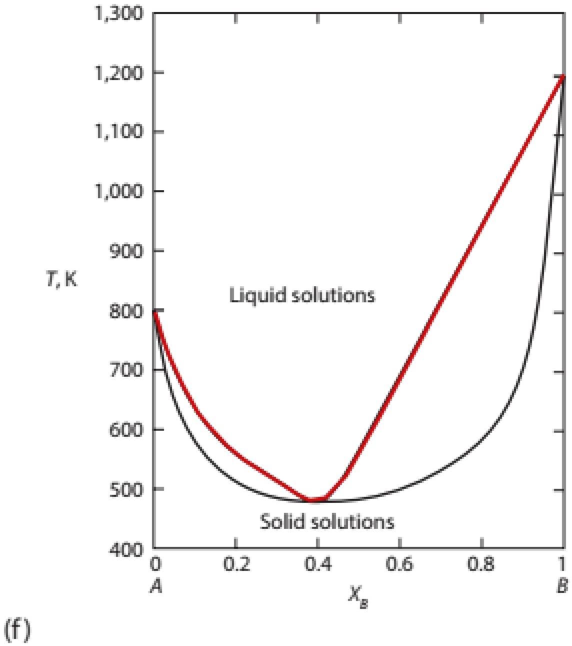


Fig 10.24

a_l = –20 kJ and α_s = 0

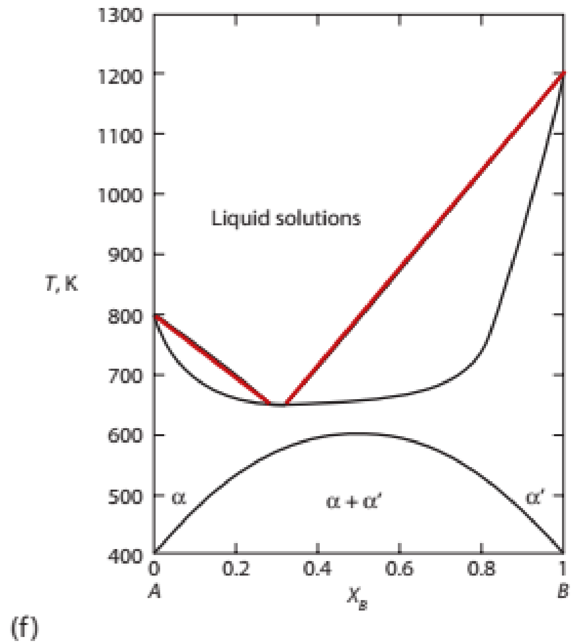


Fig 10.25

a_l = –2 kJ and α_s = 10 kJ

T_{cr}=601 K

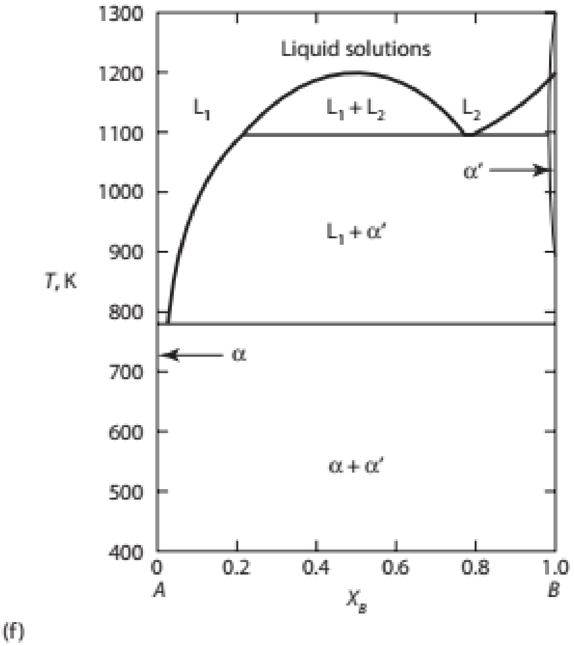


Fig 10.26

a_l = 20 kJ and α_s = 30 kJ

T_{cr}= 1803 K

10.8 the Phase Diagrams of Binary Syste
Regular Solution Behavior in the Liquid a

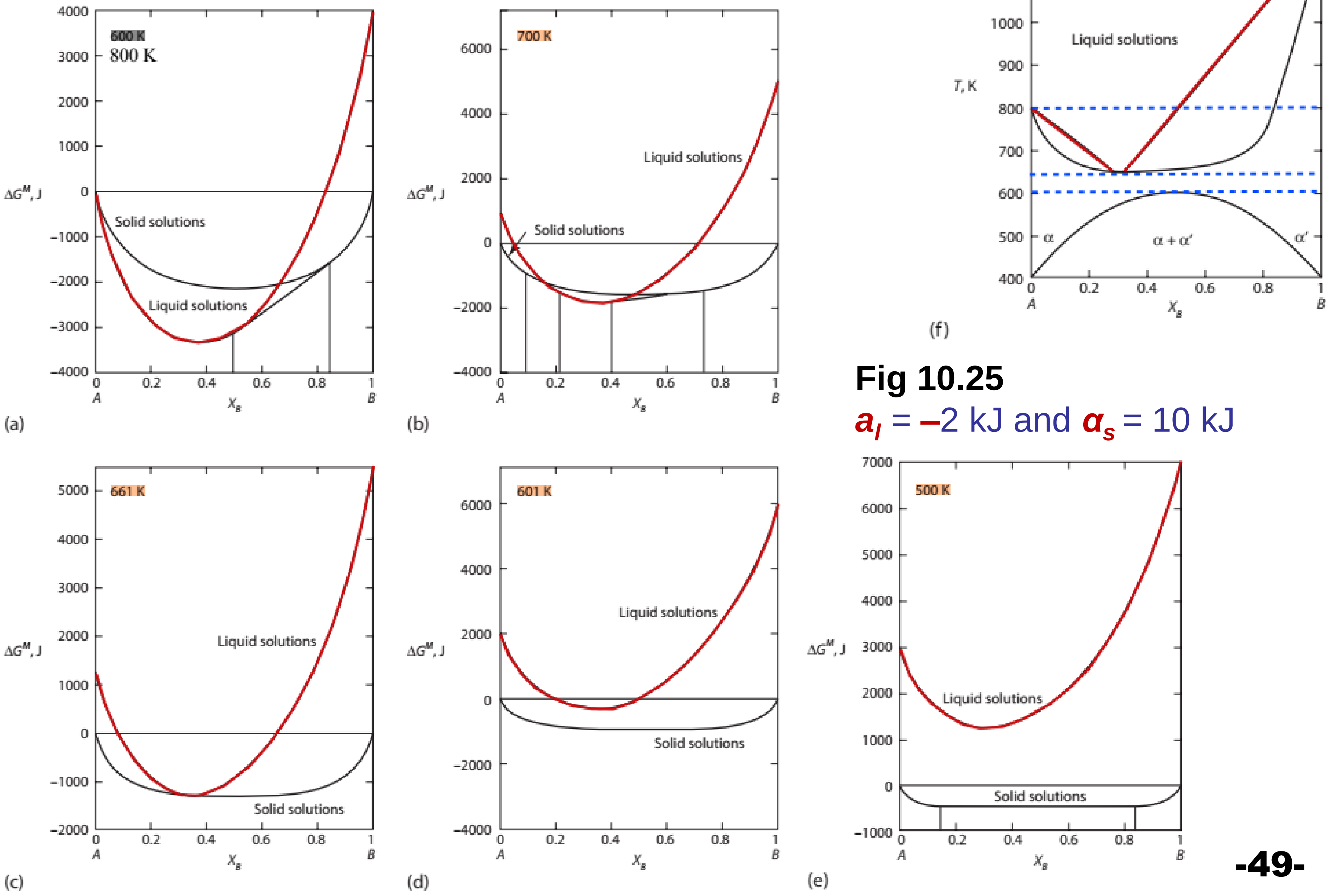


Fig 10.25

$a_l = -2$ kJ and $\alpha_s = 10$ kJ

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

- ✓ The behavior of the system in which α_l = –2000 J and α_s = 10,000 J is shown in Figure 10.25.
- ✓ As seen in Figures 10.25a–c, the behavior is similar to that shown in Figure 10.24.
- ✓ However, with a **positive** value of α_s, a critical temperature exists below which **immiscibility** occurs in the solid state, as discussed in Section 10.2.
- ✓ With α_s = 10,000 J, the **critical temperature** is 10,000/2R = **601 K**, and the mixing curves at 601 K are shown in Figure 10.25d.

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

- ✓ The phase diagram is shown in Figure 10.25f.
- ✓ With increasingly **negative** values of α_l and increasingly **positive** values of α_s , the temperature of the point of contact of the liquidus curve with the solidus curve decreases and the critical temperature in the solid-state increases, which eventually produces a eutectic system.

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

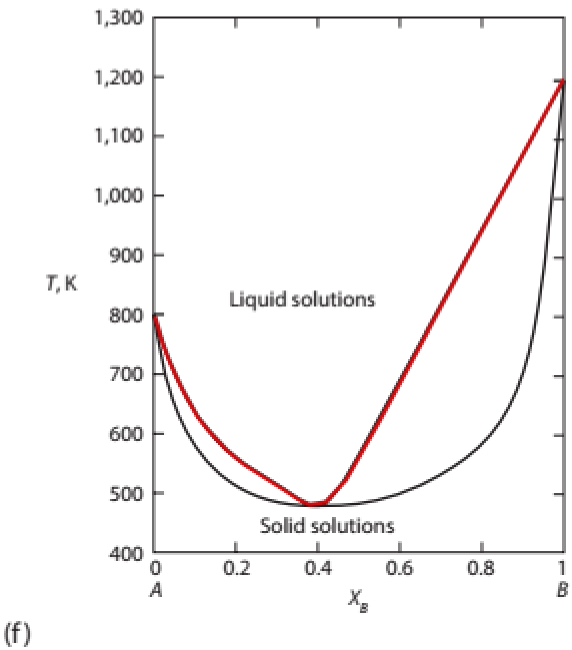


Fig 10.24

a_l = –20 kJ and α_s = 0

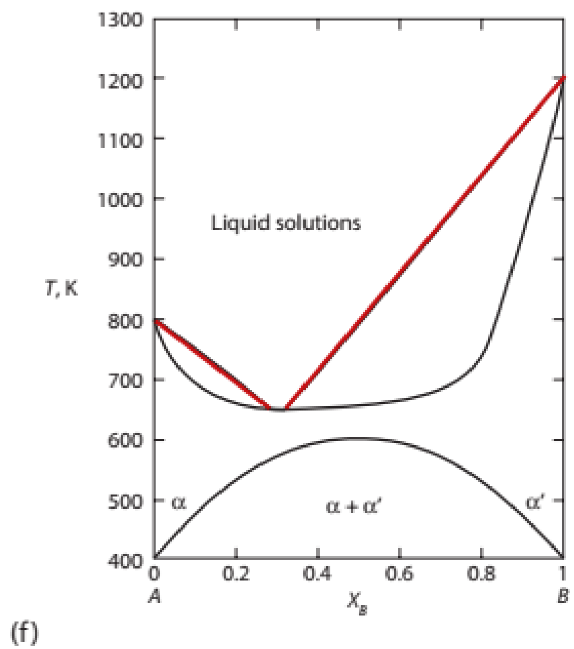


Fig 10.25

a_l = –2 kJ and α_s = 10 kJ

T_{cr}=601 K

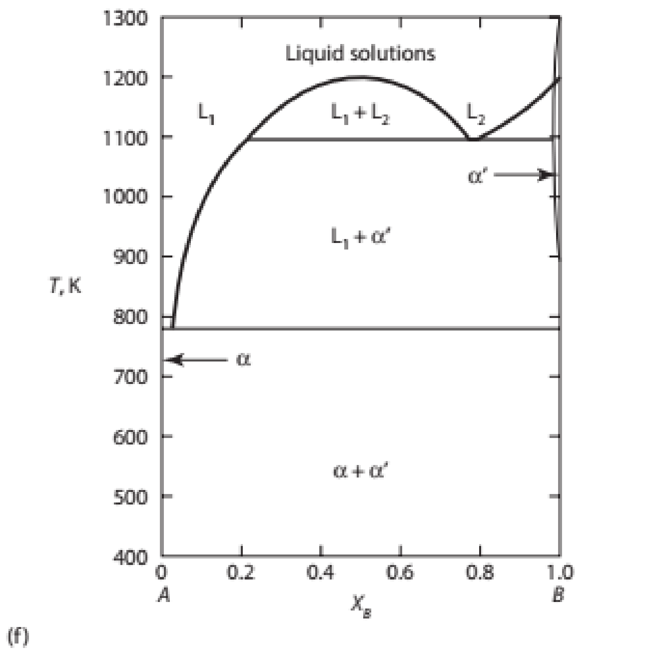
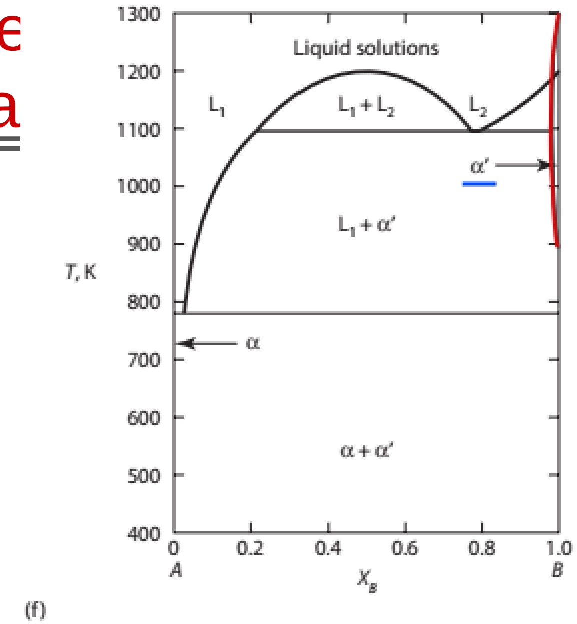


Fig 10.26

a_l = 20 kJ and α_s = 30 kJ

T_{cr}= 1803 K

10.8 the Phase Diagrams of Binary Syste
Regular Solution Behavior in the Liquid a



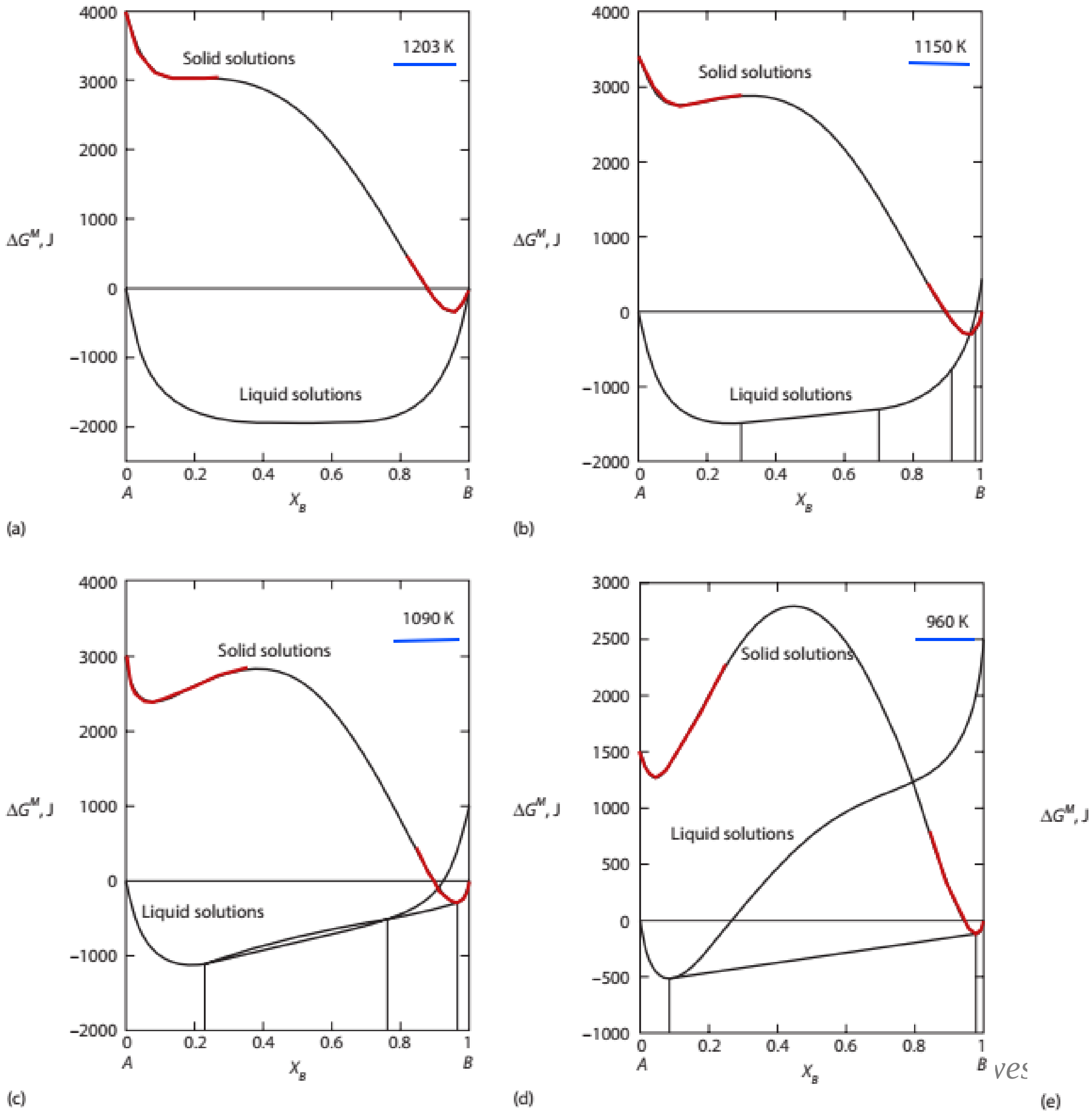
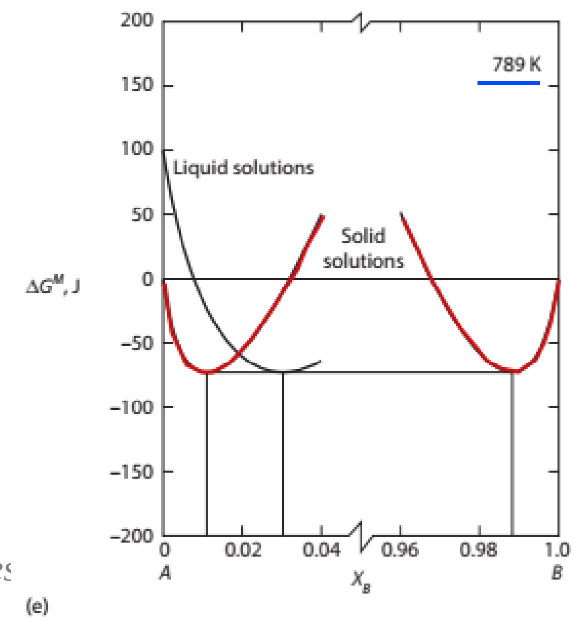


Fig 10.26

a_l =20 kJ and α_s =30 kJ



10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

α_l = **20,000** J and α_s = **30,000** J

- ✓ The T_{cr} for the liquid and solid solutions are, respectively, 1203 and 1804 K, and the ΔG^M curves at 1203, 1150, 1090, K are shown in Figure 10.26a~e.
- ✓ At $T \geq$ **1203** K, homogeneous liquids are the stable state, and at temperatures lower than 1203 K, immiscibility occurs in the **liquid** state.
- ✓ At 1150 K, two common tangents, one joining the conjugate liquid solutions L_1 **and** L_2 and one connecting the liquidus L_2 with the **solidus** α'.

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

$\boldsymbol{\alpha_l}$ = **20,000** J and $\boldsymbol{\alpha_s}$ = **30,000** J

- ✓ With decreasing temperature, the compositions of the conjugate liquid L_2 and the liquidus L_2 approach one another until, at **1090 K**, the two common tangents merge to form a triple tangent to liquid compositions at X_B = 0.23 and 0.77 and α' at X_B = 0.98. This is a monotectic equilibrium (see also Figure 10.23).

- ✓ Further cooling produces a common tangent between L_1 and α', as shown in Figure 10.26c, and at **789** K another "triple" common tangent occurs between α at X_B = 0.01, L_1 at X_B = 0.03, and α' = 0.99.

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

- ✓ At temperatures lower than the eutectic temperature of **789 K**, the liquid phase is **not stable**, and, depending on its composition,
- ✓ the system exists as ***α, α + α′***, or ***α′***.
- ✓ The monotectic and eutectic equilibria are shown in the phase diagram in Figure 10.26f.

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

✓ The influence of systematic changes in the values of α_l and α_s on the phase relationships which occur in the binary system *A–B* which forms regular solid and liquid solutions is shown in Figure 10.27.*

- In moving from the bottom of any column to the top, the value of α_s becomes more positive at constant α_l , and in moving from left to right along any row, the value of α_l becomes more positive at constant α_s .
- In the sequence Figure 10.27a–e, the liquid solutions become increasing less stable relative to the solid phases, with the consequence that the eutectic temperature is increased.
- In the transition Figure 10.27d–e, the A-liquidus becomes unstable and a monotectic equilibrium occurs.

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

- In the sequence Figure 10.27h–i, the temperature at which the three-phase equilibrium occurs is increased from 633 to 799 K, with the consequence that the eutectic equilibrium in Figure 10.23h becomes a peritectic equilibrium in Figure 10.27i.
- In Figure 10.27j, the immiscibility in the solid state disappears at a temperature below that at which a peritectic equilibrium could occur.
- With α_l = 20 kJ in Figure 10.27j, liquid immiscibility occurs at temperatures lower than 20,000 / (2 $\times$ 8.3144) = 1202 K, and hence, a monotectic equilibrium occurs at 1190 K.

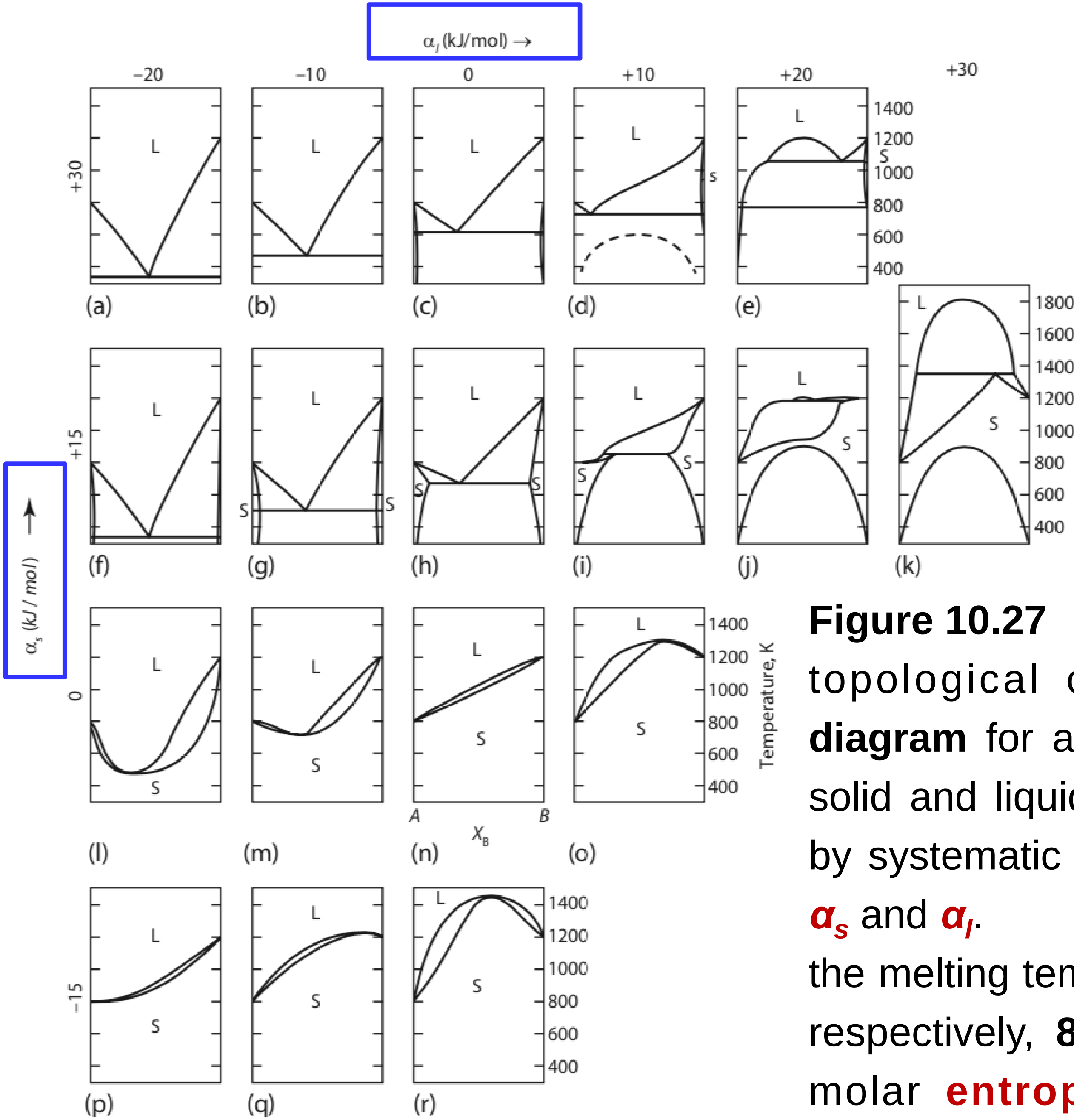
10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

- In Figure 10.27k, the three-phase L_1–L_2–α equilibrium occurs at 1360 K, which, being higher than $T_{m,(B)}$, produces a syntectic equilibrium in which the composition of the α phase lies between the compositions of the two liquids.
- In the sequence p→l→f in Figure 10.27, the solid phase becomes increasingly less stable than the liquid phase, which deepens the depression of the liquidus and solidus curves and eventually forms a eutectic.

10.8 the Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

- Figures 10.24 and 10.26, respectively, show the Gibbs free energy relations in Figure 10.27l and e.
- Figure 10.25 shows the phase equilibria occurring between those in Figure 10.27l and those in Figure 10.27f.

✓ Thus, even though the calculations are performed using a simple solution model, the trends of changing the regular solution constants α_l and α_s do shed light on the way in which various experimentally determined diagrams vary with respect to each other.



α_l : −20 ↔ +30

α_s : −15 ↔ +30

Figure 10.27

topological changes in the **phase diagram** for a system *A–B* with regular solid and liquid solutions, brought about by systematic **changes in** the values of α_s and α_l.

the melting temperatures of *A* and *B* are, respectively, **800** and **1200 K**, and the molar **entropies** of melting of both components are **10 J/K**.

Summary

✓ The molar Gibbs free energy of formation of binary solution *A–B* is given by

$$\Delta G^M = RT(X_A \ln a_A + X_B \ln a_B)$$

✓ For a regular solution,

$$\Delta G^M - \Delta G^{M,id} = G^{XS} = \alpha X_A X_B = \Delta H^M$$

✓ The criteria for equilibrium between the phases α and β in the binary system *A–B* are

$$\alpha_A(\text{in } \alpha) = \alpha_A(\text{in } \beta)$$

and

$$\alpha_B(\text{in } \alpha) = \alpha_B(\text{in } \beta)$$

Summary

- ✓ Common binary phase diagrams displaying invariant reactions include eutectic, eutectoid, peritectic, peritectoid, monotectic, monotectoid, and metatectic.

- ✓ Immiscibility becomes imminent in a regular solution at the critical value of $\alpha = 2$. The critical temperature, below which immiscibility occurs in a regular system, is given by $T_{cr} = \alpha / 2R$.

- ✓ In a binary system *A–B* which forms ideal liquid solutions and ideal solid solutions, the solidus is given by

$$X_{A(s)} = \frac{1 - \exp\left(-\frac{\Delta G^{\circ}_{m(B)}}{RT}\right)}{\exp\left(-\frac{\Delta G^{\circ}_{m(A)}}{RT}\right) - \exp\left(-\frac{\Delta G^{\circ}_{m(B)}}{RT}\right)}$$

Summary

and the liquidus line is given by

$$X_{A(l)} = \frac{\left[1-\exp(-\frac{\Delta G^{\circ}_{m(B)}}{RT})\right]\exp(-\frac{\Delta G^{\circ}_{m(A)}}{RT})}{\exp(-\frac{\Delta G^{\circ}_{m(A)}}{RT})-\exp(-\frac{\Delta G^{\circ}_{m(B)}}{RT})}$$

✓ In a binary system *A–B* which contains a eutectic equilibrium and in which the extent of solid solution is negligibly small, the liquidus lines are determined by the conditions

$$-\Delta G^{\circ}_{m(A)} = RT\ln X_A + \alpha(1-X_A)^2$$

Summary

and

$$\Delta G^{\circ}_{m(B)} = -RT\ln a_B \text{ (in the } B\text{- liquidus melt)}$$

- ✓ Thus, if the liquid solutions are ideal, the A-liquidus compositions are given by

$$-\Delta G^{\circ}_{m(A)} = RT\ln X_A$$

and if the liquid solutions are regular, the A-liquidus compositions are given by

$$-\Delta G^{\circ}_{m(A)} = RT\ln X_A + \alpha(1 - X_A)^2$$

Homework

1. The binary system Ge–Si contains complete solid and liquid solutions.

   The melting temperatures are $T_{m,Si} = 1685\ K$, and $T_{m,Ge} = 1210\ K$,

   and $\Delta H^{\circ}_{m,Si} = 50200\ J$.At 1200°C, the liquidus and solidus compositions

   are, respectively, $X_{Si} = 0.32$ and $X_{Si} = 0.665$. Calculate the value of

   $\Delta H^{\circ}_{m,Ge}$, assuming that

   a. The liquid solutions are ideal.

   b. The solid solutions are ideal.

   Which assumption gives the better estimate? The actual value of

   $\Delta H^{\circ}_{m,Ge}$ at $T_{m,Ge}$ is 36,900 J.

Homework

2. CaO and MgO form a simple eutectic system with limited ranges of solid solubility. The eutectic temperature is 2370°C. Assuming that the solutes in the two solid solutions obey Henry's law with $\gamma^{\circ}_{\mathrm{CaO}}$ in MgO = 12.88 and $\gamma^{\circ}_{\mathrm{MgO}}$ in CaO= 6.23 at 2300°C, calculate the solubility of CaO in MgO and the solubility of MgO in CaO at 2300°C.

The End